ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Geometrical Optics

English Medium

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

A. Plane Mirror

1. A plane mirror makes an angle of 30° with horizontal. If a vertical ray strikes the mirror, find the angle between mirror and reflected ray **[GO0290]**

(A) 30° (B) 45° (C) 60° (D) 90°

2. A plane mirror reflects a beam of light to form a real image. The incident beam is **[GO0291]**

(A) parallel (B) convergent (C) divergent (D) any one of the above

3. A person is in a room whose ceiling and two adjacent walls are mirrors. How many images are formed **[GO0292]**

(A) 5 (B) 6 (C) 7 (D) 8

4. An object is placed between two parallel mirrors. The number of images formed is **[GO0293]**

(A) 2 (B) 4 (C) 8 (D) infinite

5. Two plane mirrors are inclined at 70°. A ray incident on one mirror at angle θ, after reflection falls on the second mirror and is reflected from there parallel to the first mirror. θ is **[GO0294]**

(A) 50° (B) 45° (C) 30° (D) 55°

6. Plane mirrors A and B are kept at an angle θ with respect to each other. Light falls on A, is reflected, then falls on B and is reflected. The emergent ray is opposite to the incident direction. Then the angle θ is equal to **[GO0295]**

(A) 30° (B) 45° (C) 60° (D) 90°

7. Two plane mirrors are placed perpendicular to each other. A ray strikes one mirror and after reflection from the second mirror will be **[GO0296]**

(A) perpendicular to the original ray

(B) parallel to the original ray

(C) at 45° to the original ray

(D) can be at any angle to the original ray

8. A boy of height 1.5m with his eye level at 1.4m stands before a plane mirror of length 0.75m fixed on the wall. The height of the lower edge of the mirror above the floor is 0.8m. Then **[GO0297]**

(A) the boy will see his full image

(B) the boy can't see his hair

(C) the boy can't see his feet

(D) the boy can see neither his hair nor his feet

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9. An observer moves towards a plane mirror with a speed of 2 m/s. The speed of the image with respect to the observer is **[GO0298]**

(A) 1 m/s (B) 2 m/s (C) 4 m/s (C) 8 m/s

10. A plane mirror approaches a stationary person with acceleration 10 ms^{-2}. The acceleration of his image as seen by the person, will be **[GO0299]**

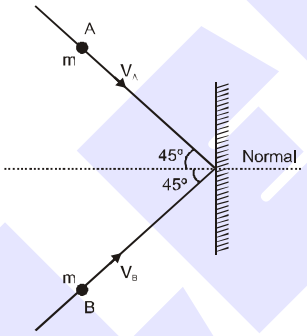
(A) 10 m/s^2 (B) 20 m/s^2 (C) 5 m/s^2 (D) can't determined

11. A plane mirror is placed at origin parallel to y-axis, facing the positive x-axis. An object starts from (2, 0, 0) with a velocity of $(2\hat{i}+2\hat{j})$ m/s. The relative velocity of image with respect to object is along

(A) positive x-axis (B) negative x-axis **[GO0300]**

(C) positive y-axis (D) negative y-aixs

12. Two bodies A and B are moving towards a plane mirror with speeds v_A and v_B respectively as shown in the figure. The speed of image of A respect to the body B is **[GO0301]**



(A) $v_A + v_B$ (B) $|\vec{v}_A + \vec{v}_B|$ (C) $\sqrt{v_A^2 + v_B^2}$ (D) $\sqrt{v_A^2 - v_B^2}$

B. Spherical Mirror

13. A straight line joining the object point and image point is always perpendicular to the mirror

(A) if mirror is plane only (B) if mirror is concave only **[GO0302]**

(C) if mirror is convex only (D) irrespective of the type of mirror.

14. The focal length of a concave mirror is 12 cm. Where should an object of height 4 cm be placed, so that a real image of 1 cm height is formed– **[GO0303]**

(A) 48 cm (B) 3 cm (C) 60 cm (D) 15 cm

15. An object is placed 10 cm in front of a convex mirror of focal length 20 cm. The distance of the image from the mirror is **[GO0304]**

(A) 10/3 cm (B) 20/3 cm (C) 10 cm (D) 40/3 cm

16. An object is placed at 20 cm from a convex mirror of focal length 20 cm. The distance of the image from the pole of the mirror is **[GO0305]**

(A) infinity (B) 10 cm (C) 15 cm (D) 40 cm

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17. A concave mirror of focal length f produces a real image n times the size of the object. The distance of the object from the mirror is **[GO0306]**

(A) $(n-1)f$ (B) $(n+1)f$ (C) $\frac{n+1}{n}f$ (D) $\frac{n-1}{n}f$

18. A concave mirror of focal length f in vacuum is placed in a medium of refractive index 2. Its focal length in the medium is **[GO0307]**

(A) f/2 (B) f (C) 2f (D) 4f

19. The image formed by a convex mirror of focal length 20 cm is half the size of the object. The distance of the object from the mirror is **[GO0308]**

(A) 10 cm (B) 20 cm (C) 30 cm (D) 40 cm

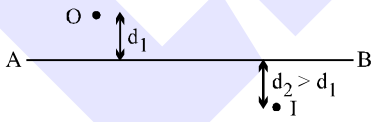
20. An object is placed in front of a convex mirror of focal length 50 cm. If the image is one-fourth the size of the object, the distance of the image from the mirror is **[GO0309]**

(A) 25 cm (B) 37.5 cm (C) 50 cm (D) 75 cm

21. An object 1 cm tall is placed 4 cm in front of a mirror. In order to produce an upright image of 3 cm height one needs a **[GO0310]**

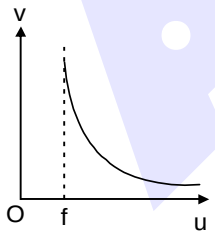
(A) convex mirror of radius of curvature 12 cm
(B) concave mirror of radius of curvature 12 cm
(C) concave mirror of radius of curvature 4 cm
(D) plane mirror of height 12 cm

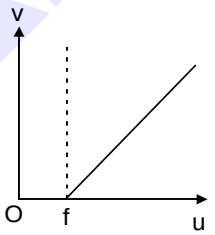
22. In the figure shown, the image of a real object is formed at point I. AB is the principal axis of the mirror. The mirror must be : **[GO0311]**

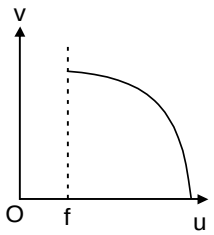


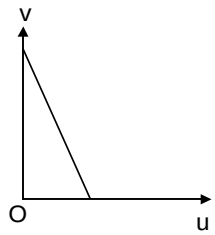
(A) concave & placed towards right I (B) concave & placed towards left of I
(C) convex and placed towards right of I (D) convex & placed towards left of I.

23. Select a graph between 'v' and 'u' for a concave mirror. **[GO0312]**

(A) 

(B) 

(C) 

(D) 

24. Check the wrong statement(s) **[GO0313]**

(A) A concave mirror can give a virtual image
(B) A concave mirror can give a diminished virtual image
(C) A convex mirror cannot give a real image
(D) A convex mirror can give a diminished virtual image

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25. If x and y be the distances of the object and image formed by a concave mirror from its focus and f be the focal length, then **[GO0314]**

(A) $xy = f$ (B) $xy = f^2$ (C) $\frac{x}{y} = f$ (D) $\frac{x}{y} = f^2$

26. A point object is placed on the principal axis at 60 cm in front of a concave mirror of focal length 40cm on the principal axis. If the object is moved with a velocity of 10 cm/s along the principal axis, then the velocity (only magnitude) of image is **[GO0315]**

(A) 20 cm/s (B) 30 cm/s (C) 40 cm/s (D) 60 cm/s

C. Refraction From Plane Surface

27. When a ray of light of frequency 6×10^{14} Hz travels from water of refractive index 4/3 to glass of refractive index 8/5, its **[GO0316]**

(A) frequency decreases by a factor of 5/6 (B) speed decreases by a factor of 5/6

(C) wavelength increases by a factor of 6/5 (D) speed increases by a factor of 6/5

28. A light wave enters from air into a medium of refractive index 1.5. The speed of light in the medium will be **[GO0317]**

(A) $2\times10^8\ m/s$ (B) $4.5\times10^8\ m/s$ (C) $3\times10^8\ m/s$ (D) $(330/1.5)\ m/s$

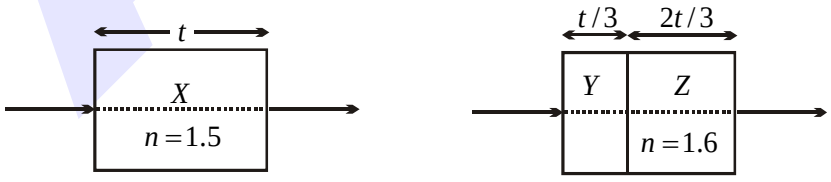
29. When light passes from one medium to another, the physical quantity that remains unchanged is

(A) velocity (B) wavelength (C) frequency (D) none **[GO0318]**

30. A beam of monochromatic blue light of wavelength 4200Å in air travels in water (n = 4/3). Its wavelength in water is **[GO0319]**

(A) 2800Å (B) 5600Å (C) 3150Å (D) 4000Å

31. Two transparent slabs have the same thickness. One is made of material X of refractive index 1.5. The other is made of two materials Y and Z having thicknesses in the ratio 1 : 2. The refractive index of Z is 1.6. If a monochromatic parallel beam passing through the slabs has the same number of wavelengths inside both, the refractive index of Y is **[GO0320]**



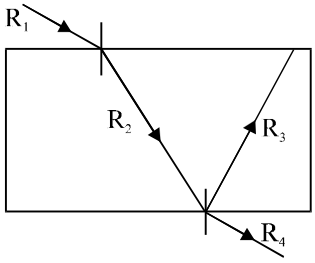
(A) 1.1 (B) 1.2 (C) 1.3 (D) 1.4

32. A ray of light passes from vacuum into a medium of refractive index μ. If the angle of incidence is twice the angle of refraction, then the angle of incidence is **[GO0321]**

(A) $\cos^{-1}(\mu/2)$ (B) $\sin^{-1}(\mu/2)$ (C) $2\cos^{-1}(\mu/2)$ (D) $2\sin^{-1}(\mu/2)$

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33. A ray R_1 is incident on the plane surface of the glass slab (kept in air) of refractive index $\sqrt{2}$ at an angle of incidence equal to the critical angle for this air glass system. The refracted ray R_2 undergoes partial reflection and refraction at the other surface. The angle between reflected ray R_3 and the refracted ray R_4 at that surface is **[GO0322]**

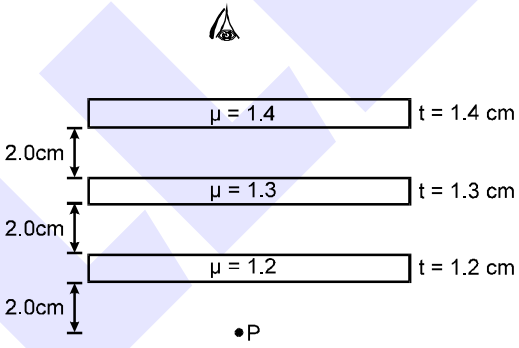


(A) 45º (B) 135º (C) 105º (D) 75º

34. A beam of light is converging towards a point on the screen. If a plane parallel plate of glass of refractive index μ and thickness *t* is introduced in the path of the beam, the convergent point is shifted **[GO0323]**

(A) $t\left[1-\frac{1}{\mu}\right]$ away (B) $t\left[1+\frac{1}{\mu}\right]$ nearer (C) $t\left[1+\frac{1}{\mu}\right]$ away (D) $t\left[1-\frac{1}{\mu}\right]$ nearer

35. Locate the image of the point P as seen by the eye in the figure. **[GO0324]**



(A) 0.9 cm below P (B) 0.8 cm above P (C) 0.7 cm above P (D) 0.9 cm above P

36. A ray of light passes from vacuum into a medium of refractive index n. If the angle of incidence is twice the angle of refraction, the angle of incidence is **[GO0325]**

(A) $\cos^{-1}(n/2)$ (B) $\sin^{-1}(n/2)$ (C) $2\cos^{-1}(n/2)$ (D) $2\sin^{-1}(n/2)$

37. How much water should be filled in a container, 21 cm in height, so that it appears half filled when viewed from the top of the container ? Given refractive index of water = 4/3 **[GO0326]**

(A) 8.0 cm (B) 10.5 cm (C) 12.0 cm (D) 14.0 cm

38. A fish in water sees an object which is 24 cm above the surface of water. The height of the object above the surface of water that will appear to the fish is **[GO0327]**

(A) 24 cm (B) 32 cm (C) 18 cm (D) 48 cm

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39. A vessel of depth d is half filled with a liquid of refractive index μ_1 and the other half is filled with a liquid of refractive index μ_2. The apparent depth of the vessel, when looked at normally, is

(A) $d(\mu_1+\mu_2)$ (B) $d\left(\frac{1}{\mu_1}+\frac{1}{\mu_2}\right)$ (C) $\frac{d}{2}(\mu_1+\mu_2)$ (D) $\frac{d}{2}\left(\frac{1}{\mu_1}+\frac{1}{\mu_2}\right)$

[GO0328]

40. A ray of light form a denser medium strikes a rarer medium at angle of incidence i. The reflected and the refracted rays make an angle of 90° with each other. The angles of reflection and refraction are r and r' respectively. The critical angle is **[GO0329]**

(A) $\sin^{-1}(\tan r)$ (B) $\sin^{-1}(\sin i)$ (C) $\sin^{-1}(\tan r')$ (D) $\tan^{-1}(\sin i)$

41. Total internal reflection of a ray of light is possible when the ray goes from **[GO0330]**

(A) denser to rarer medium and the angle of incidence is greater than the critical angle

(B) denser to rarer medium and the angle of incidence is less then the critical angle

(C) rarer to denser medium and the angle of incidence is greater than the critical angle

(D) rarer to denser medium and the angle of incidence is less then the critical angle

42. If the refractive index of water is 4/3 and that of glass is 5/3, then the critical angle of incidence for light tending to go from glass to water is **[GO0331]**

(A) $\sin^{-1}\left(\frac{3}{4}\right)$ (B) $\sin^{-1}\left(\frac{3}{5}\right)$ (C) $\sin^{-1}\left(\frac{4}{5}\right)$ (D) $\sin^{-1}\left(\frac{2}{3}\right)$

43. The critical angle of light going from medium A into medium B is θ. The speed of light in medium a is υ. The speed of light in medium B is **[GO0332]**

(A) $\frac{\upsilon}{\sin\theta}$ (B) $\upsilon\sin\theta$ (C) $\frac{\upsilon}{\tan\theta}$ (D) $\upsilon\tan\theta$

44. A fish looking up through the water sees the outside world contained in a circular horizon. If the refractive index of water is 4/3 and the fish is 12cm below the surface of water, the radius of the circle in cm is **[GO0333]**

(A) $12\times3\times\sqrt{5}$ (B) $12\times3\times\sqrt{7}$ (C) $\frac{12\times3}{\sqrt{7}}$ (D) $4\times\sqrt{5}$

45. Mirage is observed in a desert due to the phenomenon of **[GO0334]**

(A) interference (B) total reflection (C) scattering (D) double refraction

46. A diver in a lake wants to signal his distress to a person sitting on the edge of the lake flashing his water proof torch. He should direct the beam **[GO0335]**

(A) vertically upwards

(B) horizontally

(C) at an angle to the vertical which is slightly less than the critical angle

(D) at an angle to the vertical which is slightly more than the critical angle

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D. Refraction through curved surface

47. There is a small black dot at the centre C of a solid glass sphere of refractive index μ. When seen from outside, the dot will appear to be located: **[GO0336]**

(A) away from C for all values of μ

(B) at C for all values of μ

(C) at C for μ = 1.5, but away from C for μ ≠ 1.5

(D) at C only for $\sqrt{2} \le \mu \le 1.5$.

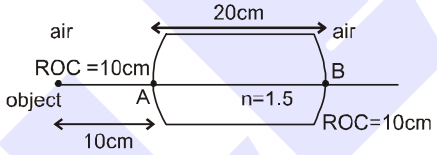
48. A point object is placed at the centre of a glass sphere of radius 6cm and refractive index 1.5. The distance of the virtual image from the surface of the sphere is :– **[GO0337]**

(A) 2cm (B) 4cm (C) 6cm (D) 12cm

49. A transparent sphere of radius R and refractive index μ is kept in air. At what distance from the surface of the sphere should a point object be placed so as to form a real image at the same distance from the other side of the sphere ? **[GO0338]**

(A) $\frac{R}{\mu}$ (B) μR (C) $\frac{R}{\mu-1}$ (D) $\frac{R}{\mu+1}$

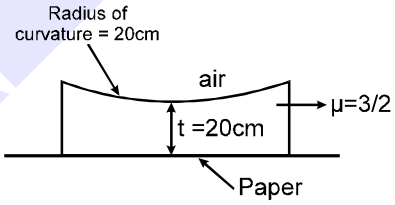
50. An object is placed 10 cm away from a glass piece (n = 1.5) of length 20 cm bound by spherical surfaces of radii of curvature 10 cm. Then the position of final image formed after two refractions at the spherical surfaces. **[GO0339]**



(A) 50 cm left of B (B) 40 cm right of B

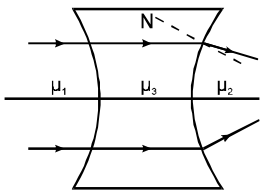
(C) 50 cm right of B (D) 60 cm right of B

51. In the given figure a plano-concave lens is placed on a paper on which a flower is drawn. How far above its actual position does the flower appear to be ? **[GO0340]**



(A) 10 cm (B) 15 cm (C) 50 cm (D) none of these

52. From the figure shown establish a relation among , μ_1, μ_2, μ_3. **[GO0341]**



(A) $\mu_3 > \mu_2 > \mu_1$ (B) $\mu_3 < \mu_2 < \mu_1$ (C) $\mu_2 > \mu_3$; $\mu_3 = \mu_1$ (D) $\mu_2 > \mu_1$; $\mu_3 = \mu_2$

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53. An object is placed 50 cm in front of a convex surface of radius 20 cm. If the surface separates air from glass of refractive index 1.5, the distance of the image from the convex surface and its nature are
(A) 30 cm, real (B) 30 cm, virtual **[GO0342]**
(C) 300 cm, real (D) 300 cm, virtual

E. Lens

54. Focal length of glass $\left(\mu = \frac{3}{2}\right)$ lens is 20 cm. Find its focal length in liquid of refractive index 2 is

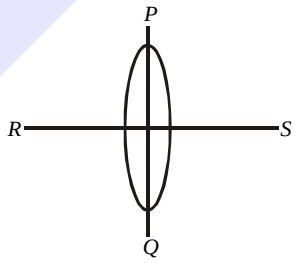
(A) 40 cm (B) 80 cm (C) –40 cm (D) 60 cm **[GO0343]**

55. A convex lens of glass has power P in air. If it is immersed in water, its power will be **[GO0344]**
(A) more than P
(B) less than P
(C) P
(D) more than P for some colours and less than P for others

56. A convex lens, made of a material of refractive index 1.5 and having a focal length of 10 cm, is immersed in a liquid of refractive index 3.0. The lens will behave as a **[GO0345]**
(A) converging lens of focal length 10 cm
(B) diverging lens of focal length 10 cm
(C) converging lens of focal length 10/3 cm
(D) diverging lens of focal length 30 cm

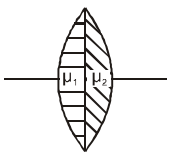
57. A convex lens of focal length 16 cm forms a real image double the size of the object. The distance of the object from the lens is **[GO0346]**
(A) 32 cm (B) 24 cm (C) 20 cm (D) 8 cm

58. The fig., shows an equiconvex lens of focal length f. If the lens is cut along PQ, the focal length of each half will be **[GO0347]**



(A) $f/2$ (B) f (C) $2f$ (D) $4f$

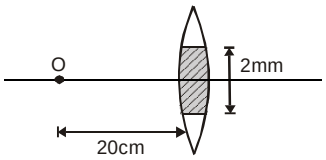
59. How many image are formed by the lens shown, if an object is kept on its axis- **[GO0348]**



(A) one (B) two (C) three (D) four

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60. A convex lens of focal length 10 cm is painted black at the middle portion as shown in figure. An object is placed at a distance of 20 cm from the lens. Then- **[GO0349]**



(A) only one image will be formed by the lens

(B) the distance between the two images formed by such a lens is 6 mm

(C) the distance between the images is 4 mm

(D) the distance between the images is 2 mm

61. An object is placed in front of a thin convex lens of focal length 30 cm and a plane mirror is placed 15cm behind the lens. If the final image of the object coincides with the object, the distance of the object from the lens is **[GO0350]**

(A) 60 cm (B) 30 cm (C) 15 cm (D) 25 cm

62. The image of an object formed by a device is always virtual and small. The device may be

(A) convex lens (B) concave mirror **[GO0351]**

(C) glass plate (D) concave lens

63. The focal length of a convex lens is f and the distance of an object from the principal focus is x. The ratio of the size of the real image to the size of the object is **[GO0352]**

(A) f/x (B) x/f (C) $\frac{f+x}{f}$ (D) $\frac{f}{f+x}$

64. If the top of a convex lens is covered with black paper, **[GO0353]**

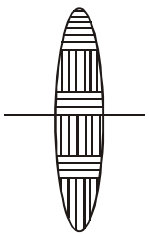
(A) the bottom half of the image will disappear

(B) the top half of the image will disappear

(C) the magnification will be reduced to half

(D) the intensity will be reduced to half

65. The layered lens as shown is made of two types of transparent materials one indicated by horizontal lines and the other by vertical lines. The number of images formed of an object will be **[GO0354]**



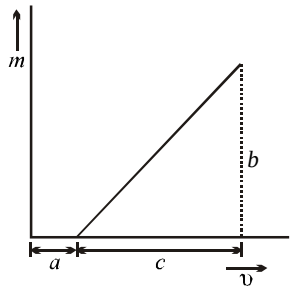
(A) 1 (B) 2 (C) 3 (D) 6

66. The maximum and minimum distances between a convex lens and an object, for the magnification of a real image to be greater than one, are **[GO0355]**

(A) $2f$ *and* f (B) f and zero (C) ∞ and $2f$ (D) $4f$ and $2f$

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67. The graph shows the variation of magnification m produced by a convex lens with the image distance υ. The focal length of the lens is **[GO0356]**

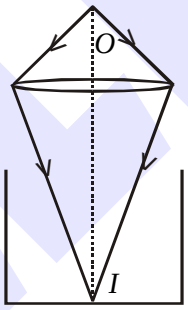


(A) b/c (B) c/b (C) a/b (D) $\frac{ab}{c}$

68. An object is placed at a distance of 12 cm from a convex lens on its principal axis and a virtual image of certain size is formed. If the object is moved 8 cm away from the lens, a real image of the same size as that of the virtual image is formed. The focal length of the lens in cm is **[GO0357]**

(A) 15 (B) 16 (C) 18 (D) 19

69. A real image of an object is formed by a convex lens at the bottom of an empty beaker. The beaker is now filled with a liquid of refractive index 1.4 to a depth of 7cm. In order to get the image again at the bottom, the beaker should be moved **[GO0358]**



(A) downward by 2 cm (B) upward by 2cm
(C) downward by 3 cm (D) upward by 3 cm

70. A 2.5 dioptre lens forms a virtual image which is 4 times the object placed perpendicularly on the principal axis of the lens. Then the required distance of the object from the lens is **[GO0359]**

(A) 35 cm (B) 40 cm (C) 30 cm (D) 25 cm

71. The distance between an object and its real image formed by a convex lens connot be **[GO0360]**

(A) greater then 2f (B) less than 2f (C) greater than 4f (D) less than 4f

72. In the displacement method, a convex lens is placed in between an object and a screen. If the magnification in the two positions are m_1 and $m_2, (m_1 > m_2)$, the distance between the two positions of the lens is x, the focal length of the lens is **[GO0361]**

(A) $\frac{x}{m_1 + m_2}$ (B) $\frac{x}{m_1 - m_2}$ (C) $\frac{x}{(m_1 + m_2)^2}$ (D) $\frac{x}{(m_1 - m_2)^2}$

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73. Two thin lenses of powers 2D and 3D are placed in contact. An object is placed at a distance of 30 cm from the combination. The distance in cm of the image from the combination is **[GO0362]**

(A) 30 (B) 40 (C) 50 (D) 60

74. A plano-convex lens of focal length 30 cm has its plane surface silvered. An object is placed 40 cm from the lens on the convex side. The distance of the image from the lens is **[GO0363]**

(A) 18 cm (B) 24 cm (C) 30 cm (D) 40 cm

75. The focal length of a convex lens is 50 cm. Its power is **[GO0364]**

(A) + 50 D (B) – 50 D (C) + 2 D (D) –2 D

76. Two lenses of power –15 D and +5 D are in contact with each other. The focal length of the combination is– **[GO0365]**

(A) –20 cm (B) –10 cm (C) +20 cm (D) +10 cm

77. A convex lens of focal length 40 cm is in contact with a concave lens of focal length 25 cm. The power of the combination in diopters is **[GO0366]**

(A) – 1.5 (B) – 6.5 (C) + 6.5 (D) + 6.67

F. Prism

78. A prism has a refractive index $\sqrt{\frac{3}{2}}$ and prism angle 90°. Find the minimum deviation produced by prism– **[GO0367]**

(A) 40° (B) 45° (C) 30° (D) 49°

79. A given ray of light suffers minimum deviation in an equilateral prism P. If refractive index increases slightly then the ray will suffer **[GO0368]**

(A) greater deviation (B) no deviation

(C) same deviation as before (D) Lesser deviation

80. If the refracting angle of a prism is 60° and the minimum deviation is 30°, then the angle of incidence is **[GO0369]**

(A) 30° (B) 60° (C) 45° (D) 90°

81. When a ray of light is refracted by a prism such that the angle of deviation is minimum, then

(A) the angle of emergence is equal to the angle of incidence **[GO0370]**

(B) the angle of emergence is greater than the angle of incidence

(C) the angle of emergence is smaller than the angle of incidence

(D) the sum of the angle of incidence and the angle of emergence is equal to 90°

82. The refracting angle of a prism is A and the refractive index of the material of the prism is $\cot(A/2)$. The angle of minimum deviation is **[GO0371]**

(A) 180° – 3A (B) 180° + 2A (C) 90° – A (D) 180° –2A

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83. If a ray of light incident normally on one face of an equilateral prism, then for no emergence from other face [$\mu \rightarrow$ refractive index of material of prism] **[GO0372]**

(A) $\mu < \frac{1}{\sqrt{3}}$ (B) $\mu = \frac{2}{\sqrt{3}}$ (C) $\mu < \frac{1}{2\sqrt{3}}$ (D) $\mu > \frac{2}{\sqrt{3}}$

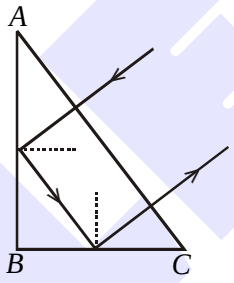
84. For an equilateral prism, it is observed that when a ray strikes grazingly at one face it emerges grazingly at the other. Its refractive index will be:- **[GO0373]**

(A) $\sqrt{3}$ (B) $\frac{2}{\sqrt{3}}$ (C) 2 (D) data not sufficient

85. A ray of monochromatic light is incident on one refracting face of a prism of angle 75^0. It passes through the prism and is incident on the other face at the critical angle. If the refractive index of the material of the prism is $\sqrt{2}$, the angle of incidence on the first face of the prism is **[GO0374]**

(A) 30^0 (B) 45^0 (C) 60^0 (D) 0^0

86. A ray falls on a prism $ABC(AB = BC)$ and travels as shown in the figure. The minimum refractive index of the prism material should be **[GO0375]**



(A) 4/3 (B) $\sqrt{2}$ (C) 3/2 (D) $\sqrt{3}$

87. The refractive index for the material of a 60° prism is 1.50. Then the angle of incidence for minimum deviation is nearly. ($\sin 42° \approx \frac{2}{3}$ and $\sin 49° \approx \frac{3}{4}$)

(A) 30° (B) 49° (C) 38° (D) 28°

GO0086

88. The refractive index of a given piece of transparent quartz is greatest for **[GO0376]**

(A) red light (B) violet light (C) green light (D) yellow light

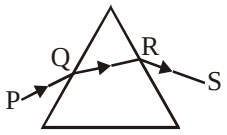
89. If f_B and f_R are the focal lengths of a convex lens for blue and red lights respectively and F_B and F_R are the respective values for a concave lens, then **[GO0377]**

(A) $f_B > f_R$ and $F_B > F_R$ (B) $f_B < f_R$ and $F_B > F_R$

(C) $f_B > f_R$ and $F_B < F_R$ (D) $f_B < f_R$ and $F_B < F_R$

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90. A ray of light is incident on an equilateral glass prism placed on a horizontal table. For minimum deviation which of the following is true ?



(A) PQ is horizontal (B) QR is horizontal

(C) RS is horizontal (D) Either PQ or RS is horizontal

GO0087

91. Chromatic aberration will be absent if for two thin lenses in contact :-

(A) $(\omega_1/F_1) + (\omega_2/F_2) = 0$ (B) $(\omega_1/F_2) + (\omega_2/F_1) = 0$

(C) $(F_1/\omega_2) + (F_2/\omega_1) = 0$ (D) $(\omega_1/\omega_2) + (F_1 + F_2) = 0$

GO0116

92. The dispersive powers of the materials of the two lenses are in the ratio 4 : 3. If the achromatic combination of these two lenses in contact is a convex lens of focal length 60 cm then the focal lengths of the component lenses are

(A) – 20 cm and 25 cm (B) 20 cm and – 25 cm

(C) – 15 cm and 40 cm (D) 15 cm and – 20 cm

GO0117

G. Defects of Eye & Optical instruments

93. A normal eye is not able to see objects closer than 25 cm because **[GO0378]**

(A) the focal length of the eye is 25 cm

(B) the distance of the retina from the eye-lens is 25 cm

(C) the eye is not able to decrease the distance between the eye-lens and the retina beyond a limit

(D) the eye is not able to decrease the focal length beyond a limit.

94. To increase the angular magnification of a simple microscope, one should increase **[GO0379]**

(A) the focal length of the lens (B) the power of the lens

(C) the aperture of the lens (D) the object size.

95. The focal length of the objective of a compound microscope is f_o and its distance from the eyepiece is L. The object is placed at a distance u from the objective. For proper working of the instrument,

(A) $L < u$ (B) $L > u$ (C) $f_o < L < 2f_o$ (D) $L > 2f_o$ **[GO0380]**

96. A compound microscope has a magnification of 30. The focal length of the eyepiece is 5 cm. If the final image is formed at the least distance of vision (25 cm), the magnification produced by the objective is **[GO0381]**

(A) 5 (B) 7.5 (C) 10 (D) 15

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97. An astronomical telescope has an angular magnification of magnitude 5 for distant objects. The separation between the objective and the eyepiece is 36 cm and the final image is formed at infinity. The focal length f_o of the objective and f_e of the eyepiece are **[GO0382]**

(A) $f_o = 45\,cm$ and $f_e = -9\,cm$ (B) $f_o = 50\,cm$ and $f_e = 10\,cm$

(C) $f_o = 7.2\,cm$ and $f_e = 5\,cm$ (D) $f_o = 30\,cm$ and $f_e = 6\,cm$

98. An astronomical telescope has an eyepiece of focal length 5 cm. If the angular magnification in normal adjustment is 10, the distance between the objective and the eye piece is **[GO0383]**

(A) 45 cm (B) 50 cm (C) 55 cm (D) 110 cm

99. In the case of hypermetropia **[GO0384]**

(A) the image of a near object is formed behind the retina

(B) the image of a distant object is formed on front of the retina

(C) a concave lens should be used for correction

(D) a convex lens should not be used for correction

100. Spherical aberration in a thin lens can be reduced by **[GO0385]**

(A) using a monochromatic light

(B) using a doublet combination

(C) using a circular annular mask over the lens

(D) increasing the size of the lens

101. Astigmatism for a human eye can be removed by using **[GO0386]**

(A) concave lens (B) convex lens

(C) cylindrical lens (D) prismatic lens

102. A person can see objects only at a distance greater than 40 cm. He is advised to use lens of power

(A) + 2.5 D (B) + 1.5 D (C) – 2.5 D (D) – 6.25 D **[GO0387]**

103. A person cannot see clearly objects at a distance less than 100 cm. The power of the spectacles required to see clearly objects at 25 cm is

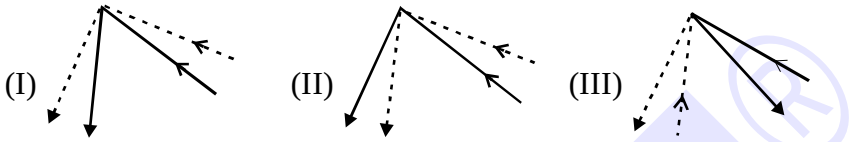
(A) + 1 D (B) + 3 D (C) + 4 D (D) + 2 D **[GO0388]**

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EXERCISE (O-2)

A. Plane Mirror

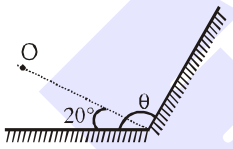
1. Each of these diagrams is supposed to show two different rays being reflected from the same point on the same mirror. Which option is correct :-



(A) Only I (B) Only II (C) Only III (D) All

GO0055

2. If θ = 110° then total number of images formed by the mirror system will be :-



(A) 2 (B) 3 (C) 4 (D) 5

GO0057

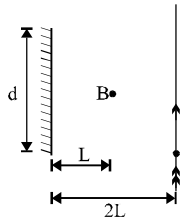
3. A point object approaches a plane mirror with a speed of 10 ms^{-1}, while the image recedes away from the mirror with a speed of 6 ms^{-1} (as seen by stationary observer). The direction and magnitude of the velocity of mirror is :-

(A) towards the object, 8 ms^{-1} (B) towards the image, 6 ms^{-1}

(C) away from the object, 8 ms^{-1} (D) away from the object, 2 ms^{-1}

GO0058

4. A point source of light *B* is placed at a distance *L* in front of the centre of a mirror of width *d* hung vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance 2*L* from it as shown. The greatest distance over which he can see the image of the light source in the mirror is :- **[IIT-JEE '2000 (Scr)]**



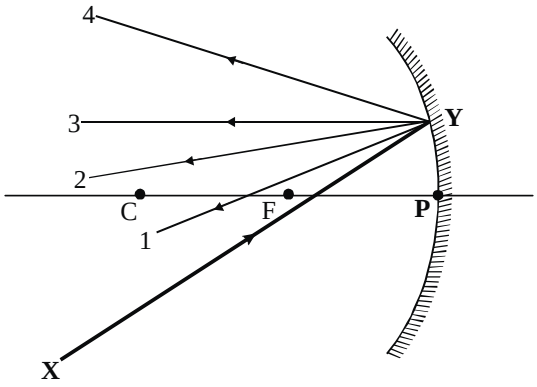
(A) d/2 (B) d (C) 2d (D) 3d

GO0061

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B. Spherical Mirror

5. Figure shows a small concave mirror with *CP* as its principal axis. A ray *XY* is incident on the mirror. Which of the four rays can be the reflected ray.



(A) 1 (B) 2 (C) 3 (D) 4

GO0062

6. The distance of an object from a spherical mirror is equal to the focal length of the mirror. Then the image:

(A) must be at infinity (B) may be at infinity

(C) may be at the focus (D) none

GO0063

7. An object is placed in front of a spherical mirror whose 2 times magnified image is formed on screen. Then choose **CORRECT** option :-

(A) Mirror is concave m = +2 (B) Mirror is concave m = –2

(C) Mirror is convex m = +2 (D) Mirror is convex m = –2

GO0064

8. A particle approaches from very large distance towards concave mirror along the principal axis. By the time the particle reaches the mirror the distance between the particle and its image

(A) first decreases then increases

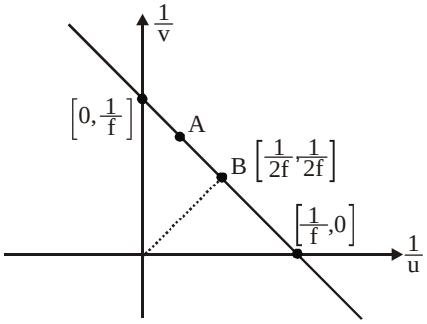
(B) first increases then decreases

(C) first increases then decreases and then again increases

(D) first decreases then increases and then again decreases

GO0066

9. Figure shows $\frac{1}{v}$ vs $\frac{1}{u}$ curve for convex mirror. Nature of image at point A is :-



(A) Real, erect, magnified (B) virtual, inverted, magnified

(C) virtual, inverted, diminished (D) virtual, erect, diminished

GO0067

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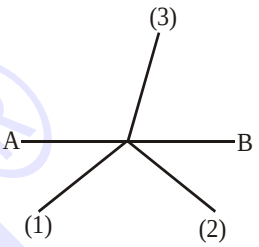
10. A car is fitted with a convex side-view mirror of focal length 20 cm. A second car 2.8 m behind the first car is overtaking the first car at a relative speed of 15 m/s. The speed of the image of the second car as seen in the mirror of the first one is:- **[AIEEE- 2011]**

(A) 10 m/s (B) 15 m/s (C) $\frac{1}{10}$ m / s (D) $\frac{1}{15}$ m / s

GO0184

C. Refraction From Plane Surface

11. AB is a boundary separating two media of different refractive indices. A ray is incident on the boundary is partially reflected and partially transmitted. Choose the **CORRECT** statement.



(A) 3 is incident ray and 1 is refracted ray

(B) 2 is incident ray and 1 is partially reflected ray

(C) 1 is incident ray and 3 is refracted ray

(D) 3 is incident ray and 2 is partially reflected ray

GO0069

12. **Statement-1:** You see a geostationary satellite above the horizon. You desire to communicate with the satellite by sending a beam of laser light. You should aim your laser slightly higher than the line of sight of the satellite.

Statement-2: Light bends away from the normal while moving from denser to rarer medium.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

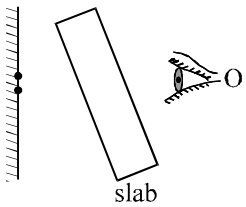
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

GO0070

13. The observer at O views two closely spaced spots on a vertical wall through an angled glass slab as shown. As seen by observer, the spots appear.



(A) shifted upward (B) shifted downward

(C) spaced farther apart (D) spaced closer together

GO0073

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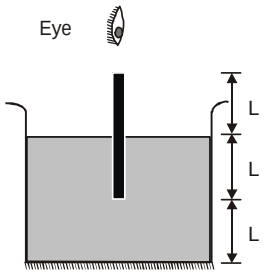
14. What is the length of the image of the rod in mirror, according to the observer in air? (refractive index of the liquid is μ)

(A) μL + L

(B) $L + \frac{L}{\mu}$

(C) $L\mu + \frac{L}{\mu}$

(D) None of these



GO0074

15. **Statement-1**: A point source of light is placed inside water. A light detector present out side, in air can detect light only in a conical region, with the apex at the source and circumscribing the circle of illuminance.

Statement-2: Ray incident from denser to rarer medium undergoes total internal reflection when their angle of incidence become more than the critical angle, this situation create a circular region from which light escapes and is called circle of illuminance.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

GO0077

16. **Statement-1**: When light falls on a sphere made of diamond total internal reflection takes place which makes it shine more than a similar sphere made of common glass.

Statement-2: Refractive index for diamond is more than refractive index of cheap glass.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

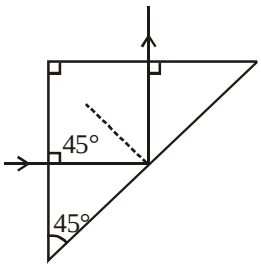
(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

GO0080

17. A light ray is incident perpendicular to one face of a 90° prism and is totally internally reflected at the glass-air interface. If the angle of reflection is 45°, we conclude that the refractive index n :

[AIEEE-2004]



(A) $n < \frac{1}{\sqrt{2}}$ (B) $n > \sqrt{2}$ (C) $n > \frac{1}{\sqrt{2}}$ (D) $n < \sqrt{2}$

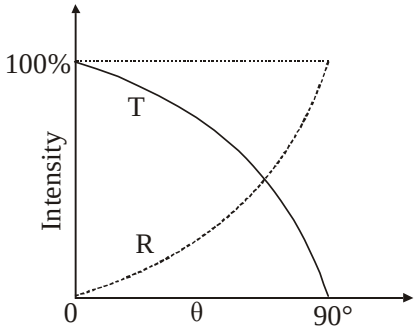
GO0081

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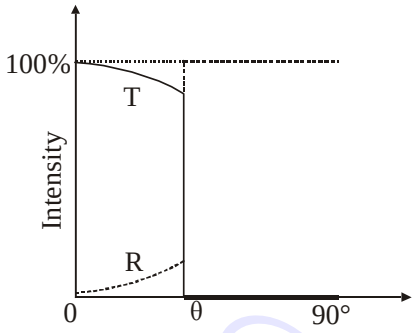
18. A light ray traveling in glass medium is incident on glass-air interface at an angle of incidence θ. The reflected (R) and transmitted (T) intensities, both as function of θ, are plotted. The correct sketch is

[IIT-JEE 2011]

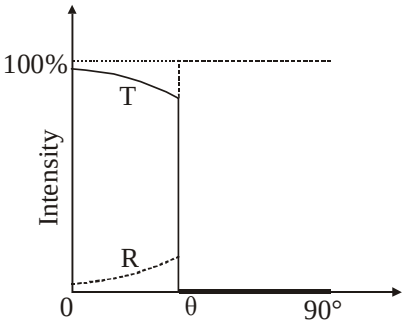
(A)



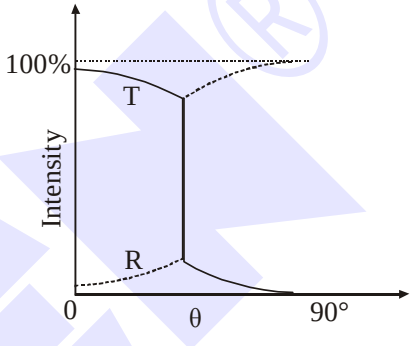
(B)



(C)



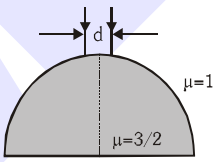
(D)



GO0202

D. Refraction through curved surface

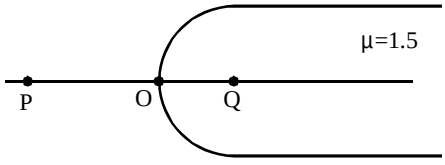
19. A beam of diameter 'd' is incident on a glass hemisphere as shown. If the radius of curvature of the hemisphere is very large in comparison to d, then the diameter of the beam at the base of the hemisphere will be :-



(A) $\frac{3}{4}$d (B) d (C) $\frac{d}{3}$ (D) $\frac{2}{3}d$

GO0093

20. One end of a glass rod of refractive index n = 1.5 is a spherical surface of radius of curvature R. The centre of the spherical surface lies inside the glass. A point object placed in air on the axis of the rod at the point P has its real image inside glass at the point Q (see fig.). A line joining the points P and Q cuts the surface at O such that OP = 2OQ. The distance PO is :-



(A) 8 R (B) 7 R (C) 2 R (D) None of these

GO0094

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21. An air bubble is inside water. The refractive index of water is 4/3. At what distance from the air bubble should a point object be placed so as to form a real image at the same distance from the bubble:

(A) 2R (B) 3R

(C) 4R (D) The air bubble cannot form a real image

GO0095

E. Lens

22. A concave lens of glass, refractive index 1.5, has both surfaces of same radius of curvature R. On immersion in a medium of refractive index 1.75, it will behave as a :-

(A) convergent lens of focal length 3.5R

(B) convergent lens of focal length 3.0 R.

(C) divergent lens of focal length 3.5 R

(D) divergent lens of focal length 3.0 R

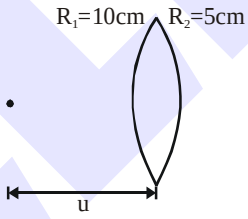
GO0097

23. A convergent (biconvex) lens is placed inside a jar filled with a liquid. The lens has focal length 20cm, when in air and its material has a refractive index of 1.5. If the liquid has a refractive index of 1.6, the focal length of the lens while in the jar, is :-

(A) –110 cm (B) –130 cm (C) –160 cm (D) –180 cm

GO0098

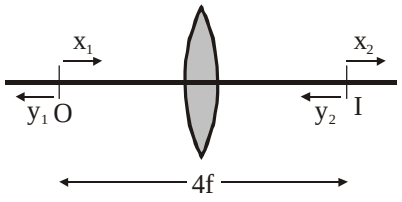
24. A thin convex lens is made of a material of refractive index 1.6. An object is kept at a distance of u from the lens on the principal axis as shown in the figure. The radius of curvature of the surfaces are 10 cm and 5cm. Now, the lens is reversed such that the face having radius of curvature 5cm lies close to the object. The difference in image position as obtained for both the cases is equal to :-



(A) 0.4 u (B) 0.6 u (C) 0.8 u (D) 0

GO0099

25. In a converging lens of focal length f and the distance between real object and its real image is 4*f*. If the object moves x_1 distance towards lens its image moves x_2 distance away from the lens and when object moves y_1 distance away from the lens its image moves y_2 distance towards the lens, then choose the correct option:-



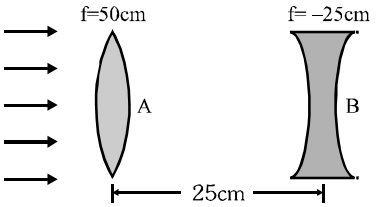
(A) $x_1 > x_2$ and $y_1 > y_2$ (B) $x_1 < x_2$ and $y_1 < y_2$

(C) $x_1 < x_2$ and $y_1 > y_2$ (D) $x_1 > x_2$ and $y_2 > y_1$

GO0100

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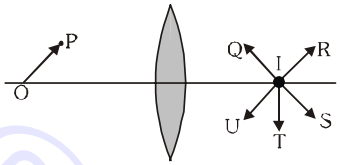
26. The two lenses shown are illuminated by a beam of parallel light from the left. Lens B is then moved slowly toward lens A. The beam emerging from lens B is :



(A) Initially parallel and then diverging
(B) Always diverging
(C) Initially converging and finally parallel
(D) Always parallel

GO0102

27. A point object O moves from the principal axis of a converging lens in a direction OP. I the image of O, will move initially in the direction :



(A) IQ (B) IR
(C) IS (D) IU

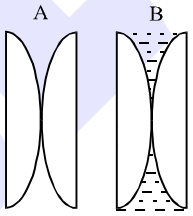
GO0103

28. When the object is at distances u_1 and u_2 the images formed by the same lens are real and virtual respectively and of the same size. Then focal length of the lens is :

(A) $\frac{1}{2}\sqrt{u_1u_2}$ (B) $\frac{u_1+u_2}{2}$ (C) $\sqrt{u_1u_2}$ (D) $\sqrt{(u_1+u_2)}$

GO0104

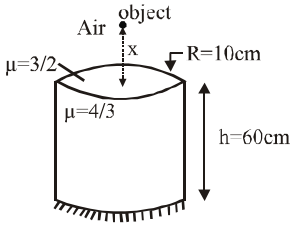
29. Figure A shows two identical plano-convex lenses in contact as shown. The combination has focal length 24 cm. Figure B shows the same with a liquid introduced between them. If refractive index of glass of the lenses is 1.50 and that of the liquid is 1.60, the focal length of the system in figure B will be :-



(A) –120 cm (B) 120 cm (C) –24 cm (D) 24 cm

GO0106

30. In the arrangement shown given that a biconvex lens of radius of curvature equal to 10 cm and concave mirror has focal length equal to 20cm. Then find the distance 'x' such that the final image formed by the system coincides with the object.



(A) 15 cm (B) 22.5 cm (C) 20 cm (D) 17.5 cm

GO0107

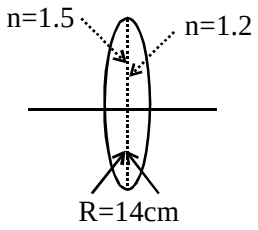
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31. A convex lens of focal length 30 cm forms an image of height 2 cm for an object situated at infinity. If a convcave lens of focal length 20 cm is placed coaxially at a distance of 26 cm in front of convex lens then size image would be :- **[IIT-JEE 2003 (Scr)]**

(A) 2.5 cm (B) 5.0 (C) 1.25 (D) None

GO0108

32. A biconvex lens is formed with two thin plano-convex lenses as shown in the figure, Refractive index n of the first lens is 1.5 and that of the second lens is 1.2. Both the curved surfaces are of the same radius of curvature R = 14 cm. For this biconvex lens, for an object distance of 40 cm, the image distance will be :- **[IIT-JEE 2012]**



(A) –280.0 cm (B) 40.0 cm (C) 21.5 cm (D) 13.3 cm

GO0204

33. A plano-convex lens of refractive index 1.5 and radius of curvature 30 cm is silvered at the curved surface. Now, this lens has been used to form the image of an object. At what distance from this lens, an object be placed in order to have a real image of the size of the object? **[AIEEE-2004]**

(A) 20 cm (B) 30 cm (C) 60 cm (D) 80 cm

GO0109

F. Prism

34. A ray of light is incident at 60° on a prism of refracting angle 30°. The emerging ray is at an angle 30° with the incident ray. The value of refractive index of the prism is :-

(A) $\frac{\sqrt{3}}{4}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\sqrt{3}$ (D) $\frac{2}{\sqrt{3}}$

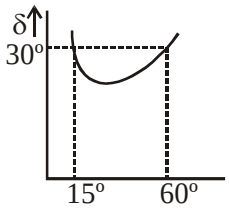
GO0082

35. The refracting angle of the prism is 60° and minimum deviation of 30°, then the angle of incidence is:-

(A) 30° (B) 45° (C) 25° (D) 60°

GO0083

36. Figure shows graph of deviation δ versus angle of incidence for a light ray striking a prism. Angle of prism is :-



(A) 30° (B) 45° (C) 60° (D) 75°

GO0084

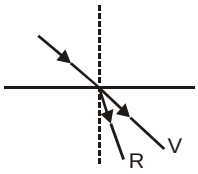
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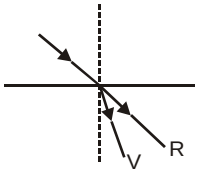
37. A beam of monochromatic light is incident at i = 50° on one face of an equilateral prism, the angle of emergence is 40°, then the angle of minimum deviation is :

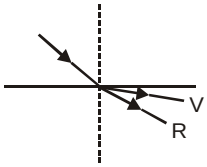
(A) 30° (B) < 30° (C) ≤ 30° (D) ≥ 30°

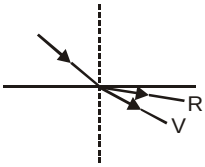
GO0091

38. A ray of light is coming from air to water. Which of the following figure shows correct dispersion of light ?

(A) 

(B) 

(C) 

(D) 

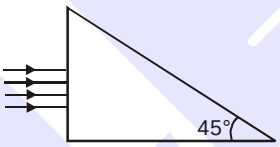
GO0111

39. The angle of prism is 5° and its refractive indices for red and violet colours are 1.5 and 1.6 respectively. The angular dispersion produced by the prism is :–

(A) 7.75° (B) 5° (C) 0.5° (D) 0.17°

GO0113

40. A beam of light consisting of red, green and blue and is incident on a right angled prism. The refractive index of the material of the prism for the above red, green and blue wavelengths are 1.39, 1.44 and 1.47 respectively. The prism will



(A) separate part of the red color from the green and blue colors
(B) separate part of the blue color from the red and green colors
(C) separate all the three colors from the other two colors
(D) not separate even partially, any colors from the other two colors

GO0114

G. Defects of Eye & Optical instruments

41. A long sighted person has a minimum distance of distinct vision of 50 cm. He wants to reduce it to 25cm. He should use a

(A) Concave lens of focal length 50 cm (B) Convex lens of focal length 25 cm
(C) Convex lens of focal length 50 cm (D) Concave lens of focal length 25 cm

GO0148

42. A person's near point is 50 cm and his far point is 3m. Powers of the lenses he required for (i) for reading (ii) for viewing distinct stars

(A) – 2D and 0.33 D (B) 2D and – 0.33 D (C) – 2D and 3D (D) 2D and – 3 D

GO0149

43. When length of a compound microscope tube increase, its magnifying power

(A) decreases (B) increases
(C) does not change (D) may increase or decrease

GO0153

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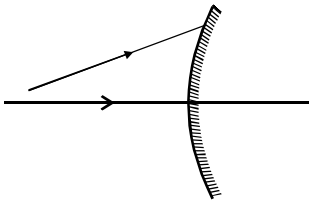
44. When the length of an astronomical telescope tube increases its magnifying power
(A) Decreases (B) Increases
(C) Does not change (D) May increase or decrease

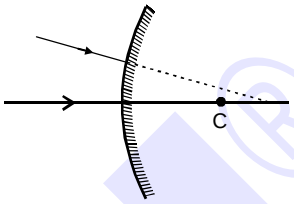
GO0155

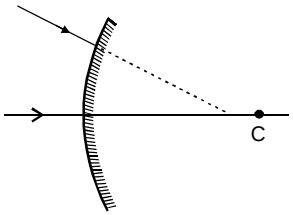
MULTIPLE CORRECT TYPE QUESTIONS

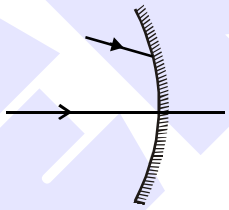
B. Spherical Mirror

45. In which of the following diagrams the image formed is virtual and inverted ?

(A) 

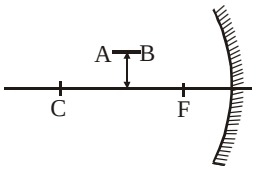
(B) 

(C) 

(D) 

GO0122

46. An object AB is placed parallel and close to the optical axis between focus F and centre of curvature C of a converging mirror of focal length f as shown in figure.



(A) Image of A will be closer than that of B from the mirror.
(B) Image of AB will be parallel to the optical axis.
(C) Length of image is equal to AB.
(D) Length of image is more than AB.

GO0123

47. A concave mirror forms an image of the sun at a distance of 12 cm from it.
(A) the radius of curvature of this mirror is 6 cm.
(B) to use it as a shaving mirror, it can be held at a distance of 8-10 cm from the face
(C) if an object is kept at a distance of 24 cm from it, the image formed will be of the same size as the object
(D) all the above alternatives are correct.

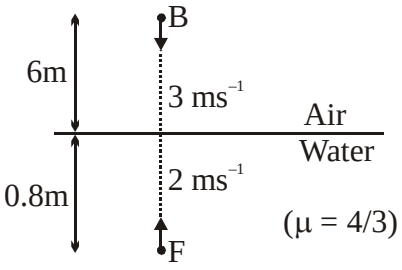
GO0125

48. The distance of a real object from the pole of a concave mirror is equal to its radius of curvature. The image must be: **[GO0389]**
(A) real (B) inverted (C) same sized (D) erect

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C. Refraction From Plane Surface

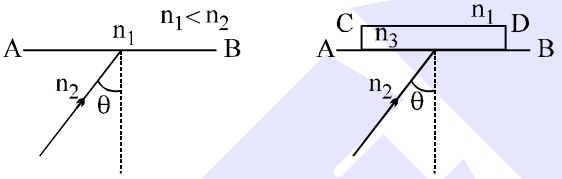
49. A fish, F in the pond, is at a depth of 0.8 m from water surface and is moving vertically upwards with velocity 2 ms^{-1}. At the same instant, a bird B is at a height of 6 m from water surface and is moving downwards with velocity 3 ms^{-1}. At this instant both are on the same vertical lines as shown in the figure. Which of the following statement(s) is(are) correct?



(A) Height of B, observed by F (from itself) is equal to 8.00 m.
(B) Depth of F, observed by B (from itself) is equal to 6.60 m.
(C) Velocity of B, observed by F (relative to itself) is equal to 5.00 ms^{-1}.
(D) Velocity of F, observed by B (relative to itself) if equal to 4.50 ms^{-1}.

GO0127

50. In the figure (a) light is incident at an angle θ which is slightly greater than the critical angle. Now, keeping the incident fixed a slab of refractive index n_3 is placed on surface AB in fig. (b). Which of the following statements are correct:

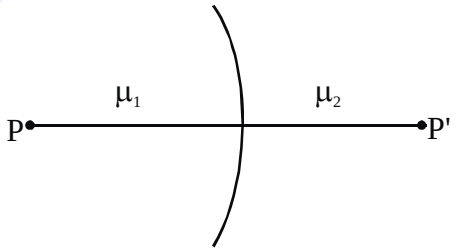


(A) total internal reflection occurs at AB for $n_3 < n_1$
(B) total internal reflection occurs at AB for $n_3 > n_1$
(C) the ray will return back to the same medium for all values of n_3
(D) total internal reflection occurs at CD for $n_3 < n_1$

GO0128

D. Refraction through curved surface

51. Two refracting media are separated by a spherical interface as shown in the figure. *PP'* is the principal axis, μ_1 and μ_2 are the refractive indices of medium of incidence and medium of refraction respectively, then



(A) if $\mu_2 > \mu_1$, then there cannot be a real image of real object.
(B) if $\mu_2 > \mu_1$, then there cannot be a real image of virtual object.
(C) if $\mu_1 > \mu_2$, then there cannot be a virtual image of virtual object.
(D) if $\mu_1 > \mu_2$, then there cannot be a real image of real object.

GO0130

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E. Lens

52. A lens is formed by a material having refecting index $\frac{3}{2}$ & radii of curvature 20cm & 10 cm. Then choose the correct option(s) :-

(A) If lens is concavo-convex & light is falling on the surface having radius of curvature 20 cm, then focal length is 40 cm

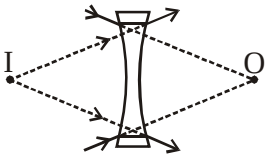
(B) If lens is concavo-convex & light is falling on the surface having radius of curvature 10 cm, then focal length is –40 cm

(C) If lens is convexo-concave & light is falling on the surface having radius of curvature 20 cm, then focal length is 40 cm

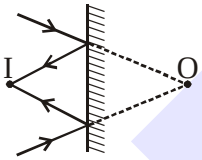
(D) If lens is convexo-concave & light is falling on the surface having radius of curvature 10 cm, then focal length is –40 cm

GO0134

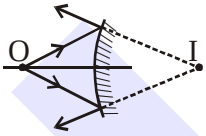
53. The nature of object and image given with each of the optical condition is shown. Choose the **correct** option(s) :-

(A) Virtual image 

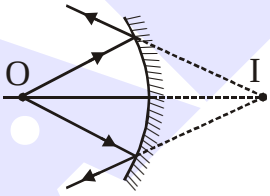
Virtual object

(B) Virtual image 

Real object

(C) Real object 

Virtual image

(D) Real object 

Virtual image

GO0121

54. Optical axis of a thin equi-convex lens is the X-axis. The co-ordinate of a point object and its image are (–20 cm, 1 cm) and (25 cm, –2 cm) respectively

(A) the lens is located at x = 5 cm
(B) the lens is located at x = –5 cm
(C) the focal length of the lens is 10 cm
(D) the focal length of the lens is 15 cm

GO0131

55. The principal axis of an optical device is along y = –1, image of a small body placed at (–30,3) is formed at a point (60, –3). Then the optical device is

(A) A convex lens of focal length 20 cm
(B) A concave mirror of focal length 60 cm
(C) A concave lens of focal length 20 cm
(D) A convex mirror of focal length 60 cm

GO0132

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56. In displacement method, the distance between object and screen is 96 cm. The ratio of length of two images formed by a convex lens placed between them is 4.84 :-

(A) Ratio of the length of object to the length of shorter image is 11/5.

(B) Distance between the two positions of the lens is 36 cm.

(C) Focal length of the lens is 20.625 cm.

(D) Distance of the lens from the shorter image is 30 cm.

GO0133

57. Which of the following silvered lenses kept in air may form real image of a real object.

(A)  (B)  (C)  (D) 

GO0135

58. A man wanted to get a picture of a Zebra. He photographed a white donkey after fitting a glass with black streaks onto the objective of his camera.

(A) the image will look like a white donkey on the photograph.

(B) the image will look like a Zebra on the photograph.

(C) the image will be more intense compared to the case in which no such glass is used.

(D) the image will be less intense compared to the case in which no such glass is used.

GO0136

F. Prism

59. For the refraction of light through a prism

(A) For every angle of deviation there are two angles of incidence.

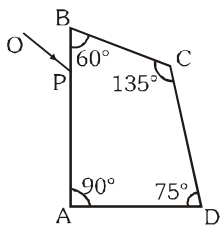
(B) The light travelling inside an equilateral prism is necessarily parallel to the base when prism is set for minimum deviation.

(C) There are two angles of incidence for maximum deviation.

(D) Angle of minimum deviation will increase if refractive index of prism is increased keeping the outside medium unchanged if $\mu_p > \mu_s$.

GO0129

60. A ray OP of monochromatic light is incident on the face AB of prism ABCD near vertex B at an incident angle of 60° (see figure). If the refractive index of the material of the prism is $\sqrt{3}$, which of the following is (are) correct? **[IIT-JEE 2010]**



(A) The ray gets totally internally reflected at face CD

(B) The ray comes out through face AD

(C) The angle between the incident ray and the emergent ray is 90°

(D) The angle between the incident ray and the emergent ray is 120°

GO0197

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COMPREHENSION TYPE QUESTIONS

B. Spherical Mirror

Paragraph for Question No. 61 to 63

A real object is placed perpendicular to the principal axis of a concave mirror at a position (1) such that the image formed is real with a magnification 2. The object is now shifted to another position (2) at a distance 15 cm from the position (1) and a real image is obtained with a magnification 8. Shifting the position of object from (2) to a third position (3) gives an image with magnification 1/2.

61. Position of the image when the object is at position (3) is :-

(A) between centre of curvature and ∞ (B) between focus and centre of curvature

(C) at focus (D) between pole and focus

GO0137

62. Focal length of the mirror is :-

(A) – 60 cm (B) –40 cm (C) –30 cm (D) –25 cm

GO0137

63. Distance of position (1) of the object from pole and that of position (2) of object from pole are, respectively :-

(A) 60 cm, 75 cm (B) 75 cm, 60 cm (C) 60 cm, 45 cm (D) 45 cm, 30 cm

GO0137

C. Refraction From Plane Surface

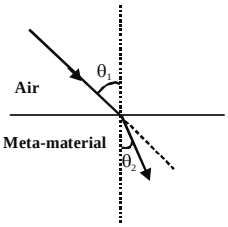
Paragraph for Questions 64 and 65

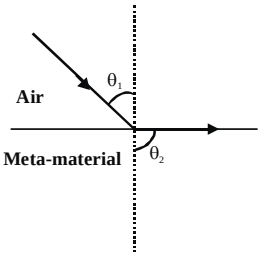
Most materials have the refractive index, n>1. So, when a light ray from air enters a naturally occurring material, then by Snell's law, $\frac{\sin\theta_1}{\sin\theta_2}=\frac{n_2}{n_1}$, it is understood that the refracted ray bends towards the normal. But it never emerges on the same side of the normal as the incident ray. According to electromagnetism, the refractive index of the medium is given by the relation, $n=\left(\frac{c}{v}\right)=\pm\sqrt{\varepsilon_r\mu_r}$, where c is the speed of electromagnetic waves in vacuum, v its speed in the medium, ε_r and μ_r are the relative permittivity and permeability of the medium respectively.

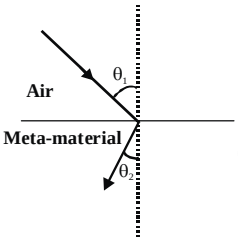
In normal materials, both ε_r and μ_r are positive, implying positive n for the medium. When both ε_r and μ_r are negative, one must choose the negative root of n. Such negative refractive index materials can now be artificially prepared and are called meta-materials. They exhibit significantly different optical behaviour, without violating any physical laws. Since n is negative, it results in a change in the direction of propagation of the refracted light. However, similar to normal materials, the frequency of light remains unchanged upon refraction even in meta-materials. **[IIT-JEE-2012]**

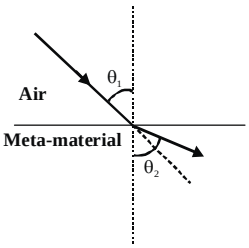
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64. For light incident from air on a meta-material, the appropriate ray diagram is

(A) 

(B) 

(C) 

(D) 

GO0205

65. Choose the correct statement. **[IIT-JEE 2012]**

(A) The speed of light in the meta-material is v = c|n|

(B) The speed of light in the meta-material is v = $\frac{c}{|n|}$

(C) The speed of light in the meta-material is v = c.

(D) The wavelength of the light in the meta-material (λ_m) is given by $\lambda_m = \lambda_{air}|n|$, where λ_{air} is the wavelength of the light in air.

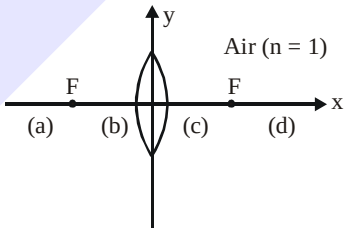
GO0205

E. Lens

Paragraph for Question no. 66 and 67

This question concerns a symmetrical lens shown, along with its two focal points. It is made of plastic with (n = 1.2), and has a focal length f. Four different regions are shown :

Here (a) $-\infty < x < -f$; (b) $-f < x < 0$; (c) $0 < x < f$; (d) $f < x < \infty$



66. If an object is placed somewhere in region (a), in which region does the image appear ?

(A) (a) (B) (b) (C) (c) (D) (d)

GO0139

67. If incident rays are converging then in which region does the image appear ?

(A) (a) (B) (b) (C) (c) (D) (d)

GO0139

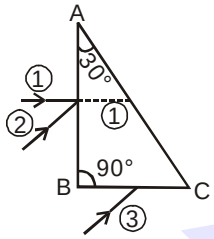
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F. Prism

Paragraph for Question No. 68 to 70

ABC is a right-angled prism kept in air. A ray (1) is incident on the face AB along the normal. Refractive index of the material of prism is the minimum value that will be required so that ray (1) undergoes total internal reflection at the face AC.

Another ray (2) is incident on the face AB such that it emerges from face AC along the normal to AC. A third ray (3) falls on the face BC and emerges from face AC such that its angle of emergence is the same as that of incidence. Assuming light (1), (2) and (3) to have the same frequency, answer the following questions.



68. Refractive index of the material of prism is :-

(A) $\sqrt{2}$ (B) 2 (C) 1.5 (D) 2.2

GO0138

69. Angle of incidence of ray (2) is :-

(A) 0° (B) 45° (C) 60° (D) 90°

GO0138

70. Deviation suffered by ray (3) is :-

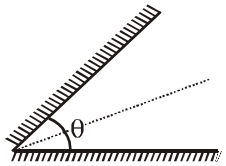
(A) 60° (B) 30° (C) 90° (D) 120°

GO0138

MATRIX MATCH TYPE QUESTIONS

A. Plane Mirror

71. Angle between two mirrors 'θ' and location of object is given in column I and some possible number of images are given in column II.



	Column–I		**Column–II**
(A)	θ = 30° and object is on bisector	(P)	9
(B)	θ = 36° and object is on bisector	(Q)	11
(C)	θ = 22.5° and object is on bisector	(R)	8
(D)	θ = 50° and object is on bisector	(S)	15

GO0140

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B. Spherical Mirror

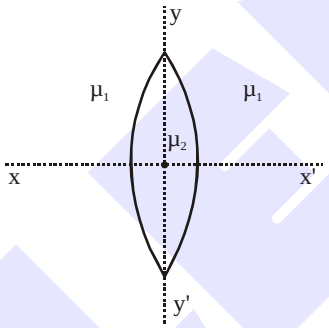
72.

	Column-I		**Column-II**
(A)	An object is placed at a distance equal to focal length from pole before convex mirror	(P)	Maginification is (∞)
(B)	An object is placed at focus before a concave mirror	(Q)	Magnification is (0.5)
(C)	An object is placed at the centre of curvature before a concave mirror.	(R)	Magnification is (1/3)
(D)	An object is placed at a distance equal to radius of curvature before a convex mirror.	(S)	Magnification is (–1)

GO0142

E. Lens

73. An equi-convex lens of refractive index μ_2 and focal length f(in air) is kept in medium of refractive index μ_1.



	Column A		**Column B**
(P)	If lens is cut in two equal parts by a plane yy'. ($\mu_1 = 1$)	(1)	Lens will act as a glass slab
(Q)	If lens is cut in two equal parts by a plane xx' . ($\mu_1 = 1$)	(2)	Lens will be converging and focal length will change.
(R)	If $\mu_1 = \mu_2$.	(3)	Lens will be converging and focal length will remain same
(S)	If $\mu_1 > \mu_2$.	(4)	Lens will be diverging and focal length will change.

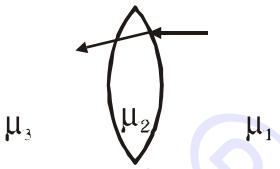
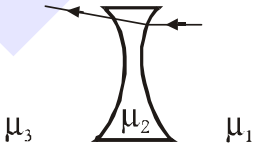
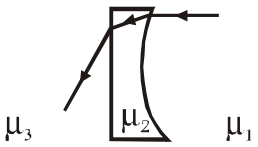
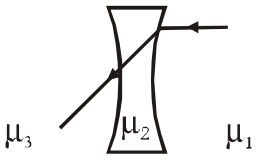
Codes :

	P	Q	R	S
(A)	2	4	1	3
(B)	2	3	1	4
(C)	3	4	2	1
(D)	1	2	3	4

GO0144

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74. Two transparent media of refractive indices μ_1 and μ_3 have a solid lens shaped transparent material of refractive index μ_2 between them as shown in figures in **Column II**. A ray traversing these media is also shown in the figures. In **Column I** different relationships between μ_1, μ_2 and μ_3 are given. Match them to the ray diagrams shown in **Column II**. **[IIT-JEE 2010]**

Column I	**Column II**
(A) $\mu_1 < \mu_2$	(P) 
(B) $\mu_1 > \mu_2$	(Q) 
(C) $\mu_2 = \mu_3$	(R) 
(D) $\mu_2 > \mu_3$	(S) 
	(T) 

GO0198

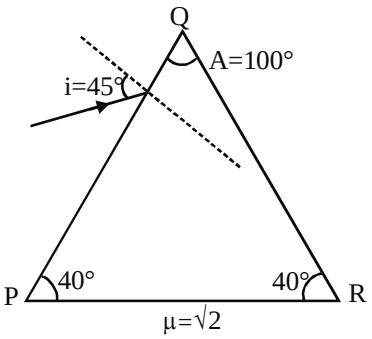
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F. Prism

75. Light is incident at surface PQ of prism as shown in column-I then match the column-I with column-II (surrounding medium is air in all cases)

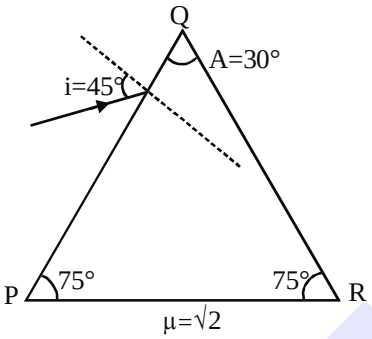
Column-I **Column-II**

(A)



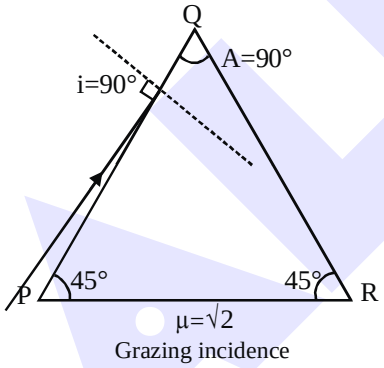
(P) Total internal reflection takes place at surface QR.

(B)



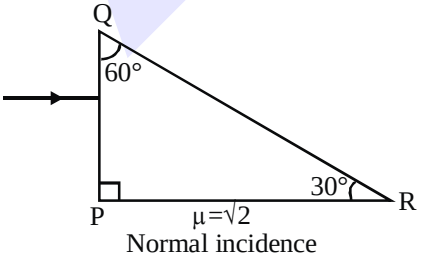
(Q) Light emerges normally from the surface QR

(C)



(R) Light emerges parallel to surface QR

(D)



(S) The light ray emerges from face PR perpendicularly

(T) When light ray passes through the prism it is parallel to the base PR.

GO0143

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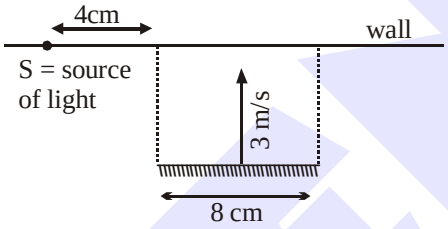
EXERCISE (S)

A. Plane Mirror

1. A fluorescent lamp of length 1 m is placed horizontally at a depth of 1.2 m below a ceiling. A plane mirror of length 0.6 m is placed below the lamp parallel to and symmetric to the lamp at a distance 2.4m from it. The length of the reflected patch of length on the ceiling _____ .

GO0001

2. A plane mirror of length 8 cm is moving with speed 3 m/s towards a wall in situation as shown in figure. Size of spot formed on the wall is 32/x cm. Find the value of x.

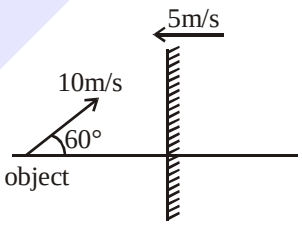


GO0002

3. There is a point object and a plane mirror. If the mirror is moved by 2 cm away from the object find the distance (in cm) which the image will move. **GO0390**

4. The angle between the velocity vector of object and image is $\frac{\pi}{\alpha}$. Velocity are shown in the figure.

Then the value of α is.



GO0003

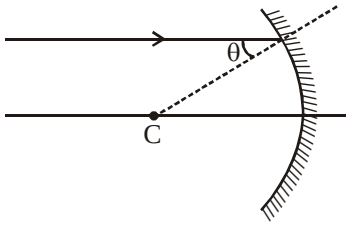
B. Spherical Mirror

5. An experimentalist devises a method for finding the radius of curvature of a convex mirror. He uses a plane mirror strip between the object and the convex mirror and adjusts it till the two virtual images formed by reflection at both the mirrors coincide without parallax. In his observations, the object distance from the convex mirror is 0.5 m while it is 0.30 m in front of the plane mirror. Find the radius of curvature (in cm) of the convex mirror.

GO0005

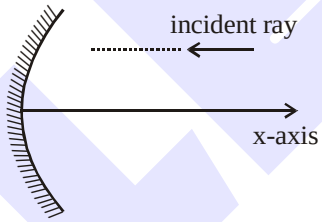
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6. A ray incident parallel to principal axis on a concave mirror at an angle θ after two reflection, reflected ray becomes parallel to incident ray. If $\theta = \frac{\pi}{N}$ (in radian) then find the value of N.



GO0009

7. A parabolic reflecting surface given by $x = \frac{y^2}{2}$, is placed at oirign, as shown. An incidnet ray moving along y = 1 is incident on it. The ray gets reflected by the surface twice. The deviation suffered by the ray is $\frac{\pi}{n}$ radians. Find n.

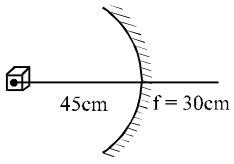


GO0010

8. The radius of curvature of a convex spherical mirror is 1.2 m. An object of height 1.2 cm is placed in front of mirror, if the distance between its virtual image and the mirror is 0.35 m? What is the height (in mm) of the image? **GO0391**

9. A thin rod of length d/3 is placed along the principal axis of a concave mirror of focal length = d such that its image, which is real and elongated, just touches the rod. Find the length of the image? **GO0006**

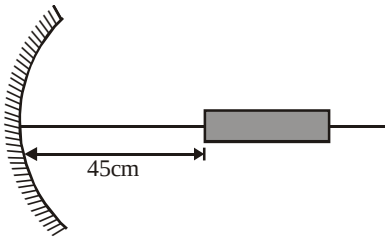
10. A cube of side length 1mm is placed on the axis of a concave mirror at a distance of 45 cm from the pole as shown in the figure. One edge of the cube is parallel to the axis. The focal length of the mirror is 30 cm. Find approximate volume of the image.



GO0007

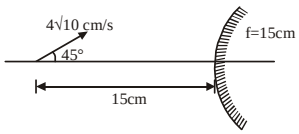
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11. An object of length 30 cm is placed on principal axis of a concave mirror of focal length 30 cm. Its one end at a distance of 45 cm as shown. If length of image is 10x (in cm) find the value of x.



GO0008

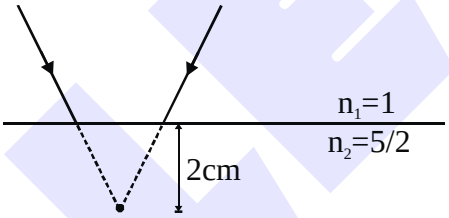
12. In the figure shown , the speed cm/s of image with respect to mirror is :



GO0011

C. Refraction From Plane Surface

13. In the given figure rays incident on an interface would converge 2 cm below the interface if they continued to move in straight lines without bending. But due to refraction, the rays will bend and meet somewhere else. Find the distance of meeting point of refracted rays below the interface (in cm). (Assuming the rays to be making small angles with the normal to the interface)



GO0012

14. An opaque cylindrical tank with an open top has a diameter of 3.00 m and is completely filled with water. When the setting sun reaches an angle of 37° above the horizon, sunlight ceases to illuminate any part of the bottom of the tank. How deep is tank in meter?

GO0014

15. A cylindrical bucket of depth 60 cm is partly filled with a liquid of refractive index 1.5 and with oil (on top of liquid) of refractive index 2. It appears that the volume of air, volume of liquid and volume of oil are equal, to an observer who views from top of the bucket. The apparent depth of the bucket as seen by the observer is given as α cm. Fill $\left(\frac{\alpha}{5}\right)$ in OMR sheet.

GO0015

16. A point object is placed 33 cm from a convex mirror of curvature radius = 40 cm. A glass plate of thickness 6 cm and index 2.0 is placed between the object and mirror, close to the mirror. Find the distance of final image from the object?

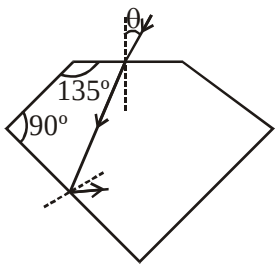
GO0013

17. A room contains air in which the speed of sound is 340 m/s. The walls of the room are made of concrete, in which the speed of sound is 1700 m/s. (a) Find the critical angle for total internal reflection of sound at the concrete–air boundary. (b) In which medium must the sound be traveling to undergo total internal reflection? (Assume sound waves follow snell's law)

GO0016

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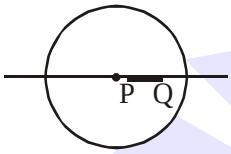
18. A ray of light enters a diamond (n = 2) from air and is being internally reflected near the bottom as shown in the figure. Find maximum value of angle θ possible ?



GO0017

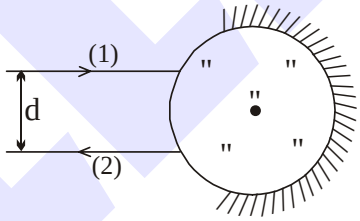
D. Refraction through curved surface

19. A small object of length 1 mm lies along the principal axis of a spherical glass of radius R = 10 cm and refractive index is 3/2. The object is seen from air along the principal axis from left. The distance of object from the centre is 5 cm. Find the size of the image. Is real, inverted ?



GO0019

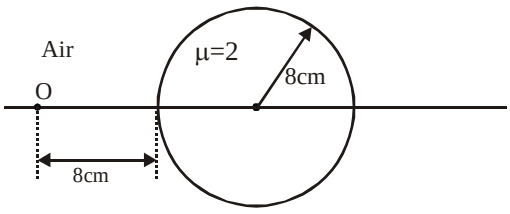
20. A transparent sphere of radius R = 2.0 m has a mirrored surface on its right half as shown in figure. A light ray travelling in air is incident on the left side of the sphere. The incident light ray(1) & exiting light ray (2) are parallel & separated by distance d = 2.0 m. Then find the refractive index of the material. (take : sin15°= 0.25)



GO0020

21. A ray incident at a point as an angle of incidence of 60° enters a glass sphere of R.I. n = $\sqrt{3}$ and is reflected and refrected at the farther surface of the sphere. The angle between the reflected and refracted rays at this surface is 10 X (in degrees), then find the value of X. **GO0392**

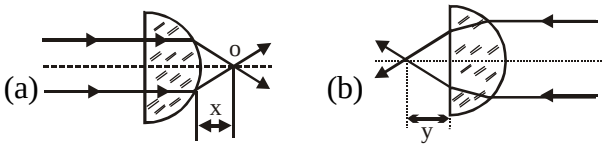
22. On the axis of a transparent sphere of refractive index (n = 2) & radius 8 cm, an object is kept at the distance 8 cm from the surface of the sphere. Find the minimum distance of the image after all possible refraction from the surface of the sphere in cm.



GO0022

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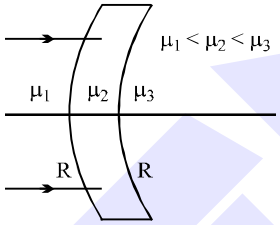
23. A narrow beam of light passing through the hemisphere of material with refractive index n, intersects at point O. Where does the beam converge (i.e. y in cm) if beam were to travel in the opposite direction as shown in figure (b)? The value of x is given to be 10 cm. Radius of the hemisphere is also 10 cm :-



GO0021

E. Lens

24. A meniscus lens is made of a material of refractive index μ_2. Both its surfaces have radii of curvature R. It has two different media of refractive indices μ_1 and μ_3 respectively, on its two sides (see figure). Calculate its focal length for $\mu_1 < \mu_2 < \mu_3$, when light is incident on it as shown. **[IIT-JEE 2003]**



GO0023

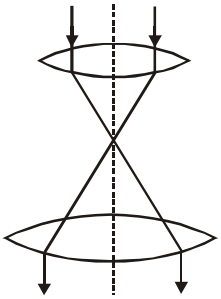
25. A plano convex lens (μ = 1.5) has a maximum thickness of 1 mm. If diameter of its aperture is 4 cm. Find (i) Radius of curvature of curved surface; (ii) its focal length in air.

GO0027

26. There are two thin symmetrical lenses, one is converging, with refractive index $n_1 = 1.70$, and the other is diverging with refractive index $n_2 = 1.50$. Both lenses have the same curvature radius of their surface equal to R = 10 cm. The lenses are put close together and submerged into water. The focal length of lens system is found to be $\frac{100}{x}$ cm in water. What is the value of x. (R.I. of water = 4/3)

GO0029

27. Consider a 'beam expander' which consists of two converging lenses of focal lengths 40 cm and 100cm having a common optical axis. A laser beam of diameter 2 mm is incident on the 40 cm focal length lens. Then what is the diameter (in mm) of the final beam (see figure)?



GO0030

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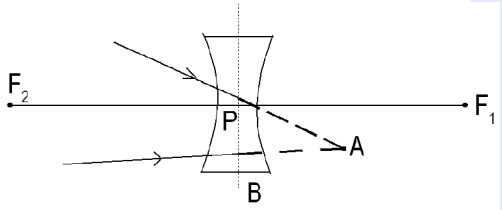
28. A lens is placed at origin, with x-axis as its principal axis. A ray of light is incident on it from the –ve side of x-axis along the line $y = \frac{x}{400} + 0.1$, where x, y are in cm. Focal length of lens is 30 cm. Find the equation of the ray after passing through the lens.

GO0034

29. A converging beam of rays is incident on a diverging lens. Having passed through the lens the rays intersect at a point 15 cm from the lens. If the lens is removed, the point where the rays meet will move 5cm closer to the mounting that holds. Find the focal length (in cm) of the lens without sign.

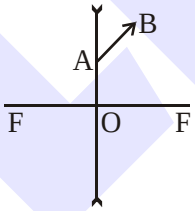
GO0024

30. The rays of a converging beam meet at a point A. A diverging lens is placed in their path in the plane B. Plot the position of the point where the rays meet after passing through the lens. The position of the principal foci FF is known.



GO0025

31. Figure shows ray AB that has passed through a divergent lens. Construct the path of the ray up to the lens if the position of its foci F is known.

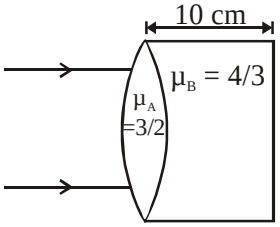


GO0026

32. A point source of light is kept at a distance of 150 cm from a converging lens, on its optical axis. The focal length of the lens is 100 cm and its diameter is 3 cm. A screen is placed on the other side of the lens, perpendicular to the axis of lens, at a distance 200 cm from it. Then find the area of the illuminated part of the screen? (Assume all rays incident on lens as paraxial)

GO0028

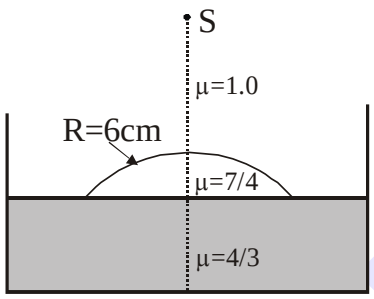
33. A parallel paraxial beam of light is incident on the arrangement as shown $\mu_A = 3/2$, $\mu_B = 4/3$, the two spherical surfaces are very close and each has radius of curvature 10 cm. Find the point where the rays are focussed. (w.r.t. point of entry)



GO0032

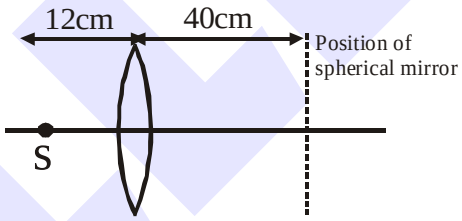
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34. Water (with refractive index = $\frac{4}{3}$) in a tank is 18 cm deep. Oil of refractive index $\frac{7}{4}$ lies on water making a convex surface of radius of curvature 'R = 6 cm' as shown. Consider oil to act as a thin lens. An object 'S' is placed 24 cm above water surface. The location of its image is at 'x' cm above the bottom of the tank. Then 'x' is **[IIT-JEE 2011]**



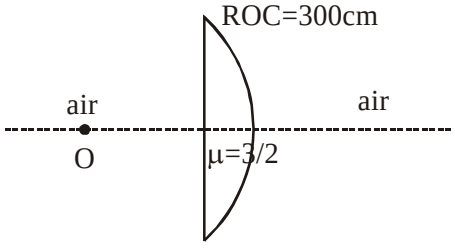
GO0203

35. A converging lens and a spherical mirror are placed with their principal axis coinciding. Their separation equals 40 cm. A point source S is placed on the principal axis at a distance of 12 cm from the lens as shown in the figure. It is found that the final beam comes out parallel to the principal axis. Focal length of the lens equals 15 cm. Find the focal length of the mirror.



GO0031

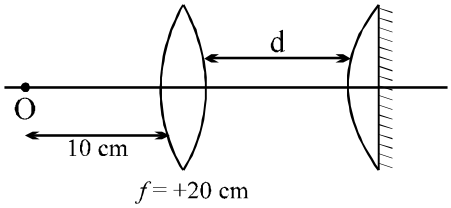
36. Plane surface of a thin planoconvex lens reflects 50% of light, while the curved surface is completely transparent, if final image of 'O' after refraction through thin lens coincides with the image formed due to partial reflection from plane surface. If distance between O and lens is x (in m) then find the value of x/4.



GO0033

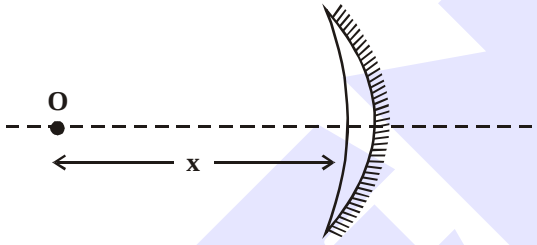
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37. A convex lens of focal length 20 cm and another plano convex lens of focal length 40 cm are placed co-axially (see fig.). The plano convex lens is silvered on plane surface. What should be the 5d (in m) so that final image of the object 'O' is formed on O itself.



GO0035

38. Radii of curvature of a concavo-convex lens (refractive index = 1.5) are 40 cm and 20 cm as shown. The convex side is silvered. The distance x (in cm) on the principal axis where an object is placed so that its image is created on the object itself, is given as 4β. Find the value of β.



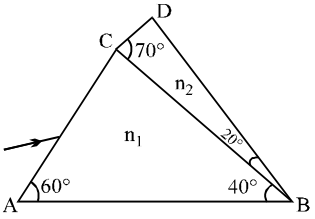
GO0036

F. Prism

39. A prism of refractive index n_1 & another prism of refractive index n_2 are stuck together without a gap as shown in the figure. The angles of the prisms are as shown. n_1 & n_2 depend on λ, the wavelength of light according to $n_1 = 1.20 + \frac{10.8\times10^4}{\lambda^2}$ & $n_2 = 1.45 + \frac{1.80\times10^4}{\lambda^2}$ where λ is in nm. **[JEE 1998]**

(i) Calculate the wavelength λ_0 for which rays incident at any angle on the interface BC pass through without bending at that interface .

(ii) For light of wavelength λ_0, find the angle of incidence i on the face AC such that the deviation produced by the combination of prisms is minimum .



GO0018

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40. A flint glass convex lens of focal length 16 cm is placed in contact with crown glass lens. The refractive indices of crown glass for violet and red colour are $\mu_v = 1.525$, $\mu_r = 1.515$ and for flint glass $\mu_{v'} = 1.655$ and $\mu_{r'} = 1.645$.

(a) Find the nature and focal length of the crown glass lens which will form an achromatic combination with the flint glass lens.

(b) What is the focal length of the combination?

GO0037

G. Defects of Eye & Optical instruments

41. A simple microscope is rated 5 X for a normal relaxed eye. What will be its magnifying power for a relaxed farsighted eye whose near point is 40 cm ? **GO0393**

42. The eyepiece of an astronomical telescope has a focal length of 10 cm. The telescope is focused for normal vision of distant objects when the tube length is 1.0 m. Find the focal length of the objective and the magnifying power of the telescope. **GO0394**

43. A Galilean telescope is 27 cm long when focused to form an image at infinity. If the objective has a focal length of 30 cm, what is the focal length of the eyepiece? **GO0395**

44. The near point and the far point of a child are at 10 cm and 100 cm. If the retina is 2.0 cm behind the eye - lens, what is the range of the power of the eye-lens ? **GO0396**

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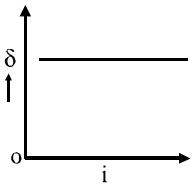
EXERCISE - (JM)

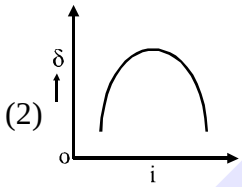
1. Diameter of a plano-convex lens is 6cm and thickness at the centre is 3 mm. If speed of light in material of lens is 2×10^8 m/s, the focal length of the lens is : **[JEE-Main- 2013]**

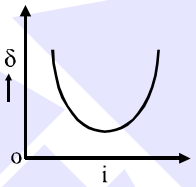
(1) 15 cm (2) 20 cm (3) 30 cm (4) 10 cm

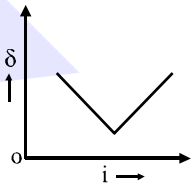
GO0188

2. The graph between angle of deviation (δ) and angle of incidence (i) for a triangular prism is represented by :- **[JEE-Main- 2013]**

(1) 

(2) 

(3) 

(4) 

GO0189

3. A thin convex lens made from crown glass $\left(\mu = \frac{3}{2}\right)$ has focal length f. When it is measured in two different liquids having refractive indices $\frac{4}{3}$ and $\frac{5}{3}$, it has the focal length f_1 and f_2 respectively. The correct relation between the focal lengths is : **[JEE-Main- 2014]**

(1) $f_2 > f$ and f_1 becomes negative (2) f_1 and f_2 both become negative

(3) $f_1 = f_2 < f$ (4) $f_1 > f$ and f_2 become negative

GO0190

4. A green light is incident from the water to the air - water interface at the critical angle (θ). Select the **correct** statement. **[JEE-Main- 2014]**

(1) The spectrum of visible light whose frequency is more than that of green light will come out to the air medium.

(2) The entire spectrum of visible light will come out of the water at various angles to the normal

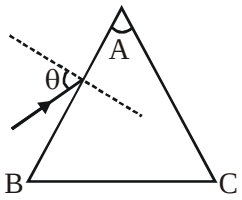
(3) The entire spectrum of visible light will come out of the water at an angle of 90° to the normal.

(4) The spectrum of visible light whose frequency is less than that of green light will come out to the air medium.

GO0191

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5. Monochromatic light is incident on a glass prism of angle A. If the refractive index of the material of the prism is μ, a ray, incident at an angle θ, on the face AB would get transmitted through the face AC of the prism provided : **[JEE-Main- 2015]**



(1) $\theta > \cos^{-1}\left[\mu\sin\left(A+\sin^{-1}\left(\frac{1}{\mu}\right)\right)\right]$

(2) $\theta < \cos^{-1}\left[\mu\sin\left(A+\sin^{-1}\left(\frac{1}{\mu}\right)\right)\right]$

(3) $\theta > \sin^{-1}\left[\mu\sin\left(A-\sin^{-1}\left(\frac{1}{\mu}\right)\right)\right]$

(4) $\theta < \sin^{-1}\left[\mu\sin\left(A-\sin^{-1}\left(\frac{1}{\mu}\right)\right)\right]$

GO0192

6. An observer looks at a distant tree of height 10 m with a telescope of magnifying power of 20. To the observer the tree appears : **[JEE-Main- 2016]**

(1) 20 times nearer (2) 10 times taller

(3) 10 times nearer (4) 20 times taller

GO0193

7. In an experiment for determination of refractive index of glass of a prism by i – δ, plot, it was found that a ray incident at angle 35°, suffers a deviation of 40° and that it emerges at angle 79°. In that case which of the following is closest to the maximum possible value of the refractive index? **[JEE-Main- 2016]**

(1) 1.8 (2) 1.5 (3) 1.6 (4) 1.7

GO0194

8. A diverging lens with magnitude of focal length 25 cm is placed at a distance of 15 cm from a converging lens of magnitude of focal length 20 cm. A beam of parallel light falls on the diverging lens. The final image formed is : **[JEE-Main- 2017]**

(1) real and at a distance of 40 cm from the divergent lens

(2) real and at a distance of 6 cm from the convergent lens

(3) real and at a distance of 40 cm from convergent lens

(4) virtual and at a distance of 40 cm from convergent lens.

GO0195

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

9. The eye can be regarded as a single refracting surface . The radius of curvature of this surface is equal to that of cornea (7.8 mm). This surface separates two media of refractive indices 1 and 1.34. Calculate the distance from the refracting surface at which a parallel beam of light will come to focus. **[JEE-Main-2019_Jan]**

(1) 2 cm (2) 1 cm (3) 3.1 cm (4) 4.0 cm

GO0250

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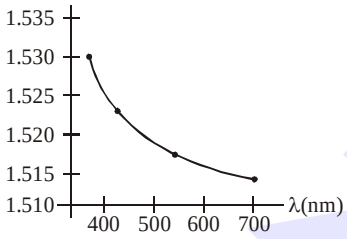
10. A plano convex lens of refractive index μ_1 and focal length f_1 is kept in contact with another plano concave lens of refractive index μ_2 and focal length f_2. If the radius of curvature of their spherical faces is R each and $f_1 = 2f_2$, then μ_1 and μ_2 are related as : **[JEE-Main-2019_Jan]**

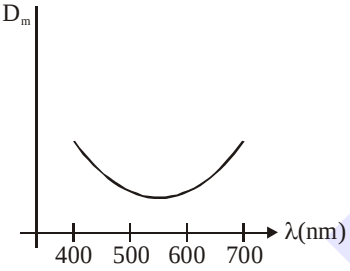
(1) $\mu_1 + \mu_2 = 3$
(2) $2\mu_1 - \mu_2 = 1$
(3) $2\mu_2 - \mu_1 = 1$
(4) $3\mu_2 - 2\mu_1 = 1$

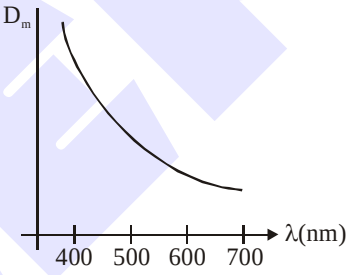
GO0251

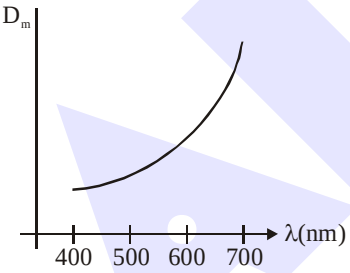
11. The variation of refractive index of a crown glass thin prism with wavelength of the incident light is shown. Which of the following graphs is the correct one, if D_m is the angle of minimum deviation?

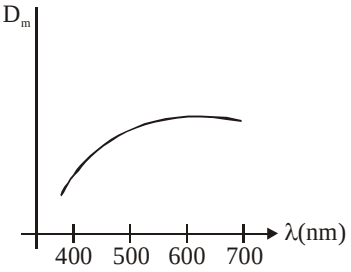
[JEE-Main-2019_Jan]



(1)


(2)


(3)


(4)


GO0252

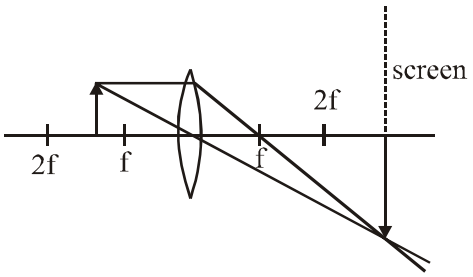
12. An object is at a distacen of 20 m from a convex lens of focal length 0.3 m. The lens forms an image of the object. If the object moves away from the lens at a speed of 5 m/s, the speed and direction of the image will be : **[JEE-Main-2019_Jan]**

(1) 0.92×10^{-3} m/s away from the lens
(2) 2.26×10^{-3} m/s away from the lens
(3) 1.16×10^{-3} m/s towards the lens
(4) 3.22×10^{-3} m/s towards the lens

GO0253

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13. Formation of real image using a biconvex lens is shown below :



If the whole set up is immersed in water without disturbing the object and the screen position, what will one observe on the screen ? **[JEE-Main-2019_Jan]**

(1) Image disappears (2) No change

(3) Erect real image (4) Magnified image

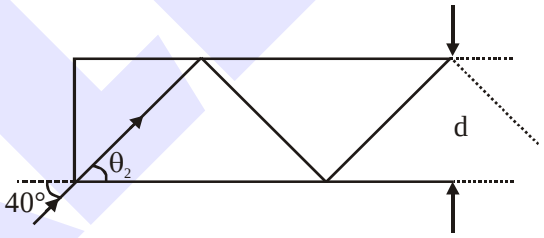
GO0254

14. A convex lens (of focal length 20 cm) and a concave mirror, having their principal axes along the same lines, are kept 80 cm apart from each other. The concave mirror is to the right of the convex lens. When an object is kept at a distance of 30 cm to the left of the convex lens, its image remains at the same position even if the concave mirror is removed. The maximum distance of the object for which this concave mirror, by itself would produce a virtual image would be :- **[JEE-Main-2019_April]**

(1) 20 cm (2) 10 cm (3) 25 cm (4) 30 cm

GO0255

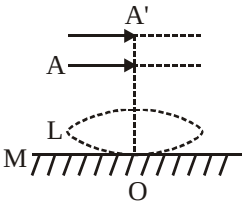
15. In figure, the optical fiber is ℓ = 2m long and has a diameter of d = 20 μm. If a ray of light is incident on one end of the fiber at angle θ_1 = 40°, the number of reflection it makes before emerging from the other end is close to: (refractive index of fibre is 1.31 and sin 40° = 0.64) **[JEE-Main-2019_April]**



(1) 55000 (2) 57000 (3) 66000 (4) 45000

GO0256

16. A thin convex lens L (refractive index = 1.5) is placed on a plane mirror M. When a pin is placed at A, such that OA = 18 cm, its real inverted image is formed at A itself, as shown in figure. When a liquid of refractive index μ_1 is put between the lens and the mirror, The pin has to be moved to A', such that OA' = 27 cm, to get its inverted real image at A' itself. The value of μ_1 will be :- **[JEE-Main-2019_April]**

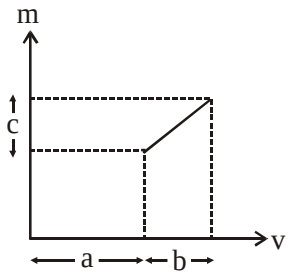


(1) $\sqrt{2}$ (2) $\frac{4}{3}$ (3) $\sqrt{3}$ (4) $\frac{3}{2}$

GO0257

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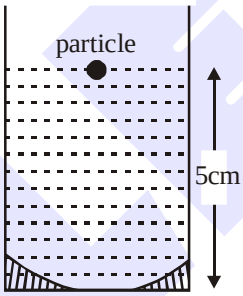
17. The graph shows how the magnification m produced by a thin lens varies with image distance v. What is the focal length of the lens used ? **[JEE-Main-2019_April]**



(1) $\frac{b^2c}{a}$ (2) $\frac{b^2}{ac}$ (3) $\frac{a}{c}$ (4) $\frac{b}{c}$

GO0258

18. A concave mirror has radius of curvature of 40 cm. It is at the bottom of a glass that has water filled up to 5 cm (see figure). If a small particle is floating on the surface of water, its image as seen, from directly above the glass, is at a distance d from the surface of water. The value of d is close to : (Refractive index of water = 1.33) **[JEE-Main-2019_April]**



(1) 8.8 cm (2) 11.7 cm (3) 6.7 cm (4) 13.4 cm

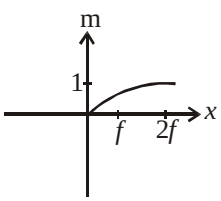
GO0259

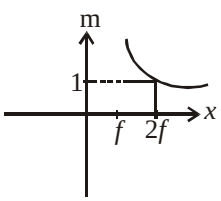
19. If we need a magnification of 375 from a compound microscope of tube length 150 mm and an objective of focal length 5 mm, the focal length of the eye-piece, should be close to : **[JEE-Main-2020_Jan]**

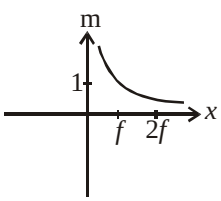
(1) 22 mm (2) 12 mm (3) 33 mm (4) 2 mm

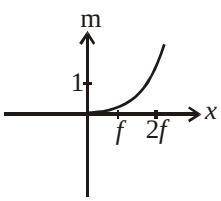
GO0260

20. An object is gradually moving away from the focal point of a concave mirror along the axis of the mirror. The graphical representation of the magnitude of linear magnification (m) versus distance of the object from the mirror (x) is correctly given by : (Graphs are drawn schematically and are not to scale) **[JEE-Main-2020_Jan]**

(1) 

(2) 

(3) 

(4) 

GO0261

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21. There is a small source of light at some depth below the surface of water (refractive index = $\frac{4}{3}$) in a tank of large cross sectional surface area. Neglecting any reflection from the bottom and absorption by water, percentage of light that emerges out of surface is (nearly) :
[Use the fact that surface area of a spherical cap of height h and radius of curvature r is 2πrh]:

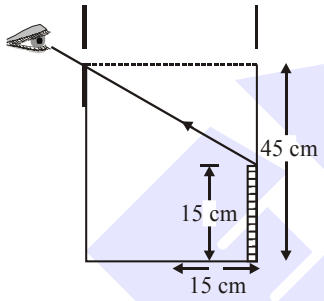
[JEE-Main-2020_Jan]

(1) 17% (2) 21% (3) 34% (4) 50%

GO0262

22. An observer can see through a small hole on the side of a jar (radius 15 cm) at a point at height of 15 cm from the bottom (see figure). The hole is at a height of 45 cm. When the jar is filled with a liquid up to a height of 30 cm the same observer can see the edge at the bottom of the jar. If the refractive index of the liquid N/100, where N is an integer, the value of N is_____.

[JEE-Main-2020_Sep]



GO0263

23. The distance between an object and a screen is 100 cm. A lens can produce real image of the object on the screen for two different positions between the screen and the object. The distance between these two positions is 40 cm. If the power of the lens is close to $\left(\frac{N}{100}\right)$D where N is an integer, the value of N is _____.

[JEE-Main-2020_Sep]

GO0264

24. In a compound microscope, the magnified virtual image is formed at a distance of 25 cm from the eye-piece. The focal length of its objective lens is 1 cm. If the magnification is 100 and the tube length of the microscope is 20 cm, then the focal length of the eye-piece lens (in cm) is _____.

[JEE-Main-2020_Sep]

GO0265

25. A short straight object of height 100 cm lies before the central axis of a spherical mirror whose focal length has absolute value |f| = 40cm. The image of object produced by the mirror is of height 25 cm and has the same orientation of the object. One may conclude from the information : **GO0397**

[JEE-Main-2021_Feb]

(1) Image is real, same side of concave mirror.

(2) Image is virtual, opposite side of concave mirror.

(3) Image is real, same side of convex mirror.

(4) Image is virtual, opposite side of convex mirror.

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26. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : For a simple microscope, the angular size of the object equals the angular size of the image.

Reason R : Magnification is achieved as the small object can be kept much closer to the eye than 25cm and hence it subtends a large angle.

In the light of the above statements, choose the most appropriate answer from the options given below : **[JEE-Main-2021_Feb]**

(1) A is true but R is false **GO0398**

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) Both A and R are true and R is the correct explanation of A

(4) A is false but R is true

27. The incident ray, reflected ray and the outward drawn normal are denoted by the unit vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively. Then choose the correct relation for these vectors. **[JEE-Main-2021_Feb]**

(1) $\vec{b} = \vec{a} + 2\vec{c}$ (2) $\vec{b} = 2\vec{a} + \vec{c}$ **GO0399**

(3) $\vec{b} = \vec{a} - 2(\vec{a}.\vec{c})\vec{c}$ (4) $\vec{b} = \vec{a} - \vec{c}$

28. A deviation of 2° is produced in the yellow ray when prism of crown and flint glass are achromatically combined. Taking dispersive powers of crown and flint glass are 0.02 and 0.03 respectively and refractive index for yellow light for these glasses are 1.5 and 1.6 respectively. The refracting angles for crown glass prism will be _____ ° (in degree) (Round off to the Nearest Integer) **GO0400**

[JEE-Main-2021_March]

29. Your friend is having eye sight problem. She is not able to see clearly a distant uniform window mesh and it appears to her as non-uniform and distorted. The doctor diagnosed the problem as : **GO0401**

[JEE-Main-2021_March]

(1) Astigmatism

(2) Myopia with Astigmatism

(3) Presbyopia with Astigmatism

(4) Myopia and hypermetropia

30. An object viewed from a near point distance of 25 cm, using a microscopic lens with magnification '6', gives an unresolved image. A resolved image is observed at infinite distance with a total magnification double the earlier using an eyepiece along with the given lens and a tube of length 0.6 m, if the focal length of the eyepiece is equal to _____ cm. **[JEE-Main-2021_July]**

GO0402

31. A ray of laser of a wavelength 630 nm is incident at an angle of 30° at the diamond-air interface. It is going from diamond to air. The refractive index of diamond is 2.42 and that of air is 1. Choose the correct option. **[JEE-Main-2021_July]**

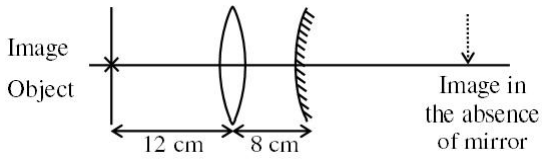
(1) angle of refraction is 24.41° (2) angle of refraction is 30° **GO0403**

(3) refraction is not possible (4) angle of refraction is 53.4°

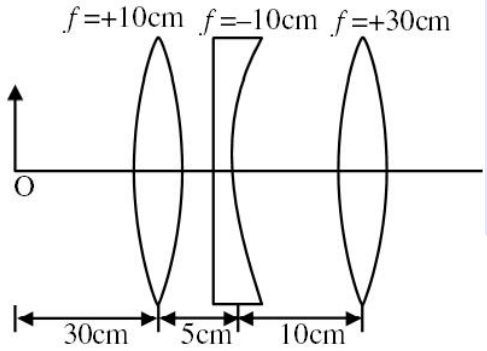
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32. An object is placed at a distance of 12 cm from a convex lens. A convex mirror of focal length 15 cm is placed on other side of lens at 8 cm as shown in the figure. Image of object coincides with the object. When the convex mirror is removed, a real and inverted image is formed at a position. The distance of the image from the object will be .....(cm) **[JEE-Main-2021_Aug]**

GO0404

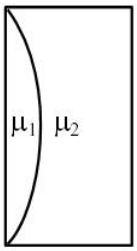


33. Find the distance of the image from object O, formed by the combination of lenses in the figure : **[JEE-Main-2021_Aug]**



(1) 75 cm (2) 10 cm (3) 20 cm (4) infinity **GO0405**

34. Curved surfaces of a plano-convex lens of refractive index μ_1 and a plano-concave lens of refractive index μ_2 have equal radius of curvature as shown in figure. Find the ratio of radius of curvature to the focal length of the combined lenses. **[JEE-Main-2021_Aug]**



(1) $\frac{1}{\mu_2 - \mu_1}$ (2) $\mu_1 - \mu_2$ (3) $\frac{1}{\mu_1 - \mu_2}$ (4) $\mu_2 - \mu_1$ **GO0406**

35. Car B overtakes another car A at a relative speed of 40 ms^{-1}. How fast will the image of car B appear to move in the mirror of focal length 10 cm fitted in car A, when the car B is 1.9 m away from the car A? **[JEE-Main-2021_Aug]**

(1) 4 ms^{-1} (2) 0.2 ms^{-1} (3) 40 ms^{-1} (4) 0.1 ms^{-1} **GO0407**

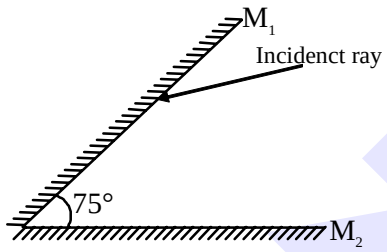
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36. A ray of light is incident at an angle of incidence 60° on the glass slab of refractive index $\sqrt{3}$. After refraction, the light ray emerges out from other parallel faces and lateral shift between incident ray and emergent ray is $4\sqrt{3}$ cm. The thickness of the glass slab is _____ cm. **GO0408**

[JEE-Main-2022_June]

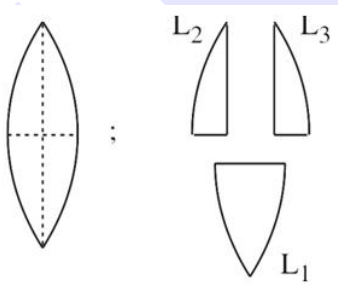
37. A light ray is incident, at an incident angle θ_1, on the system of two plane mirrors M_1 and M_2 having an inclination angle 75° between them (as shown in figure). After reflecting from mirror M_1 it gets reflected back by the mirror M_2 with an angle of reflection 30°. The total deviation of the ray will be ..... degree. **[JEE-Main-2022_June]**

GO0409



38. A convex lens has power P. It is cut into two halves along its principal axis. Further one piece (out of the two halves) is cut into two halves perpendicular to the principal axis (as shown in figure). Choose the incorrect option for the reported pieces. **[JEE-Main-2022_June]**

GO0410



(A) Power of $L_1 = \frac{P}{2}$ (B) Power of $L_2 = \frac{P}{2}$ (C) Power of $L_3 = \frac{P}{2}$ (D) Power of $L_1 = P$

39. Which of the following statement is correct ? **[JEE-Main-2022_July]**

(A) In primary rainbow, observer sees red colour on the top and violet on the bottom **GO0411**

(B) In primary rainbow, observer sees violet colour on the top and red on the bottom

(C) In primary rainbow, light wave suffers total internal reflection twice before coming out of water drops

(D) Primary rainbow is less bright than secondary rainbow.

40. For an object placed at a distance 2.4 m from a lens, a sharp focused image is observed on a screen placed at a distance 12 cm from the lens. A glass plate of refractive index 1.5 and thickness 1 cm is introduced between lens and screen such that the glass plate plane faces parallel to the screen. By what distance should the object be shifted so that a sharp focused image is observed again on the screen? **[JEE-Main-2022_July]**

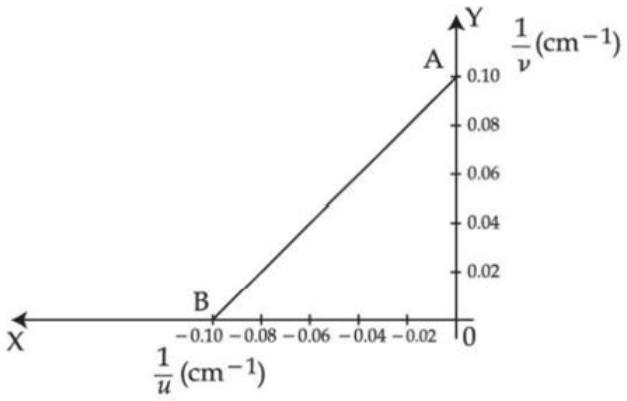
(A) 0.8 m (B) 3.2 m (C) 1.2 m (D) 5.6 m **GO0412**

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41. The graph between $\frac{1}{u}$ and $\frac{1}{v}$ for a thin convex lens in order to determine its focal length is plotted as shown in the figure. The refractive index of length is 1.5 and its both the surfaces have same radius of curvatures R. The value of R will be ____cm. (Where u = object distance , v = image distance)

GO0413

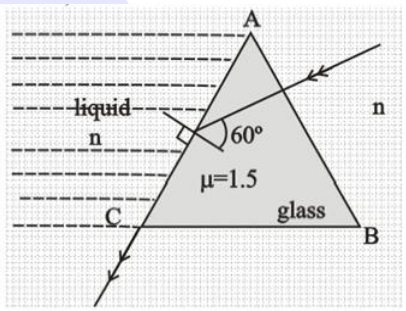
[JEE-Main-2022_July]



42. In the given figure, the face AC of the equilateral prism is immersed in a liquid of refractive index 'n'. For incident angle 60° at the side AC, the refracted light beam just grazes along face AC. The refractive index of the liquid $n=\frac{\sqrt{x}}{4}$. The value of x is_____ . (Given refractive index of glass = 1.5)

GO0414

[JEE-Main-2022_July]



43. The X-Y plane be taken as the boundary between two transparent media M_1 and M_2. M_1 in $Z \geq 0$ has a refractive index of $\sqrt{2}$ and M_2 with Z < 0 has a refractive index of $\sqrt{3}$. A ray of light travelling in M_1 along the direction given by the vector $\vec{A}=4\sqrt{3}\hat{i}-3\sqrt{3}\hat{j}-5\hat{k}$, is incident on the plane of separation. The value of difference between the angle of incident in M_1 and the angle of refraction in M_2 will be _____ degree.

[JEE-Main-2022_July]

GO0415

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EXERCISE-(JA)

1. A ray of light travelling in the direction $\frac{1}{2}\left(\hat{i}+\sqrt{3}\hat{j}\right)$ is incident on a plane mirror. After reflection, it travels along the direction $\frac{1}{2}\left(\hat{i}-\sqrt{3}\hat{j}\right)$. The angle of incidence is :- **[JEE-Advance-2013]**

(A) 30° (B) 45° (C) 60° (D) 75°

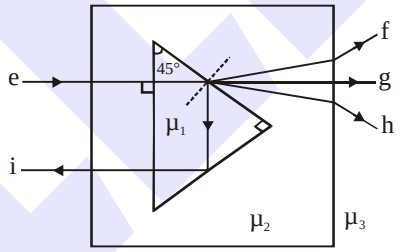
GO0206

2. The image of an object, formed by a plano-convex lens at a distance of 8 m behind the lens, is real and is one-third the size of the object. The wavelength of light inside the lens is $\frac{2}{3}$ times the wavelength in free space. The radius of the curved surface of the lens is :- **[JEE-Advance-2013]**

(A) 1 m (B) 2 m (C) 3 m (D) 6 m

GO0207

3. A right angled prism of refractive index μ_1 is placed in a rectangular block of refractive index μ_2, which is surrounded by a medium of refractive index μ_3, as shown in the figure. A ray of light 'e' enters the rectangular block at normal incidence. Depending upon the relationships between μ_1, μ_2, and μ_3, it takes one of the four possible paths 'ef', 'eg', 'eh' or 'ei'. **[JEE-Advance-2013]**



Match the paths in List I with conditions of refractive indices in List II and select the correct answer using the codes given below the lists :

	List I		**List II**
P.	$e \to f$	1.	$\mu_1 > \sqrt{2}\mu_2$
Q.	$e \to g$	2.	$\mu_2 > \mu_1$ and $\mu_2 > \mu_3$
R.	$e \to h$	3.	$\mu_1 = \mu_2$
S.	$e \to i$	4.	$\mu_2 < \mu_1 < \sqrt{2}\mu_2$ and $\mu_2 > \mu_3$

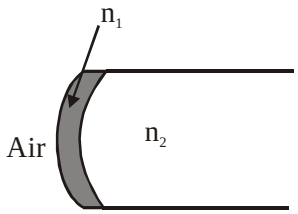
Codes :

	P	Q	R	S
(A)	2	3	1	4
(B)	1	2	4	3
(C)	4	1	2	3
(D)	2	3	4	1

GO0208

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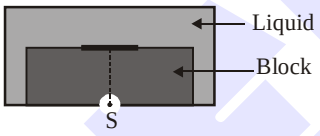
4. A transparent thin film of uniform thickness and refractive index $n_1 = 1.4$ is coated on the convex spherical surface of radius R at one end of a long solid glass cylinder of refractive index $n_2 = 1.5$, as shown in the figure. Rays of light parallel to the axis of the cylinder traversing through the film from air to glass get focused at distance f_1 from the film, while rays of light traversing from glass to air get focused at distance f_2 from the film. Then **[JEE-Advance-2014]**



(A) $|f_1| = 3R$ (B) $|f_1| = 2.8\ R$ (C) $|f_2| = 2R$ (D) $|f_2| = 1.4\ R$

GO0209

5. A point source S is placed at the bottom of a transparent block of height 10 mm and refractive index 2.72. It is immersed in a lower refractive index liquid as shown in the figure. It is found that the light emerging from the block to the liquid forms a circular bright spot of diameter 11.54 mm on the top of the block. The refractive index of the liquid is :- **[JEE-Advance-2014]**



(A) 1.21 (B) 1.30 (C) 1.36 (D) 1.42

GO0210

6. Four combinations of two thin lenses are given in List-I. The radius of curvature of all curved surfaces is r and the refractive index of all the lenses is 1.5. Match lens combinations in List-I with their focal length in List-II and select the correct answer using the code given below the lists.

[JEE-Advance-2014]

List–I		**List–II**	
(P)		(1)	2r
(Q)		(2)	$\frac{r}{2}$
(R)		(3)	–r
(S)		(4)	r

Code :

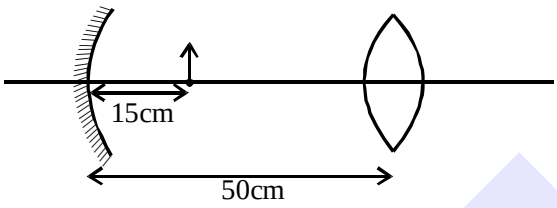
(A) P-1, Q-2, R-3, S-4 (B) P-2, Q-4, R-3, S-1

(C) P-4, Q-1, R-2, S-3 (D) P-2, Q-1, R-3, S-4

GO0211

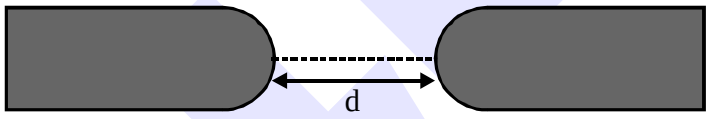
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7. Consider a concave mirror and a convex lens (refractive index = 1.5) of focal length 10 cm each, separated by a distance of 50 cm in air (refractive index = 1) as shown in the figure. An object is placed at a distance of 15 cm from the mirror. Its erect image formed by this combination has magnification M_1. When the set-up is kept in a medium of refractive index 7/6 the magnification becomes M_2. The magnitude $\left|\frac{M_2}{M_1}\right|$ is. **[JEE-Advance-2015]**



GO0212

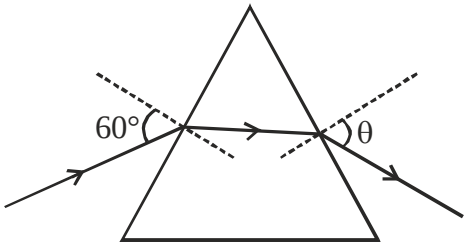
8. Two identical glass rods S_1 and S_2 (refractive index = 1.5) have one convex end of radius of curvature 10 cm. They are placed with the curved surfaces at a distance d as shown in the figure, with their axes (shown by the dashded line) aligned. When a point source of light P is placed inside rod S_1 on its axis at a distance of 50 cm from the curved face, the light rays emanating from it are found to be parallel to the axis inside S_2. The distance d is : **[JEE-Advance-2015]**



(A) 60 cm (B) 70 cm
(C) 80 cm (D) 90 cm

GO0213

9. A monochromatic beam of light is incident at 60° on one face of an equilateral prism of refractive index n and emerges from the opposite face making an angle θ(n) with the normal (see the figure). For n = $\sqrt{3}$ the value of θ is 60° and $\frac{d\theta}{dn} = m$. The value of m is **[JEE-Advance-2015]**

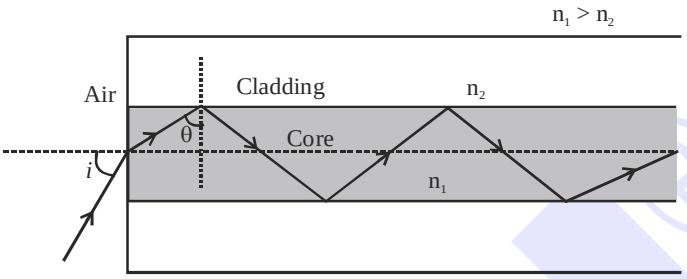


GO0214

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Paragraph for Question No. 10 and 11

Light guidance in an optical fiber can be understood by considering a structure comprising of thin solid glass cylinder of refractive index n_1 surrounded by a medium of lower refractive index n_2. The light guidance in the structure takes place due to successive total internal reflections at the interface of the media n_1 and n_2 as shown in the figure. All rays with the angle of incidence i less than a particular value of i_m are confined in the medium of refractive index n_1. The numerical aperture (NA) of the structure is defined as $\sin i_m$. **[JEE-Advance-2015]**



10. For two structures namely S_1 with $n_1 = \sqrt{45}/4$ and $n_2 = 3/2$, and S_2 with $n_1 = 8/5$ and $n_2 = 7/5$ and taking the refractive index of water to be 4/3 and that of air to be 1, the correct option(s) is (are)

(A) NA of S_1 immersed in water is the same as that of S_2 immersed in liquid of refractive index $\frac{16}{3\sqrt{15}}$.

(B) NA of S_1 immersed in liquid of refractive index $\frac{6}{\sqrt{15}}$ is the same as that of S_2 immersed in water.

(C) NA of S_1 placed in air is the same as that of S_2 immersed in liquid of refractive index $\frac{4}{\sqrt{15}}$.

(D) NA of S_1 placed in air is the same as that of S_2 placed in water.

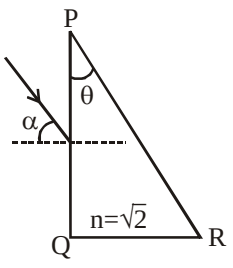
GO0215

11. If two structures of same cross-sectional area, but different numerical apertures NA_1 and NA_2 ($NA_2 < NA_1$) are joined longitudinally, the numerical aperture of the combined structure is

(A) $\frac{NA_1NA_2}{NA_1 + NA_2}$ (B) $NA_1 + NA_2$ (C) NA_1 (D) NA_2

GO0215

12. A parallel beam of light is incident from air at an angle α on the side PQ of a right angled triangular prism of refractive index $n = \sqrt{2}$. Light undergoes total internal reflection in the prism at the face PR when α has a minimum value of 45°. The angle θ of the prism is : **[JEE-Advance-2016]**



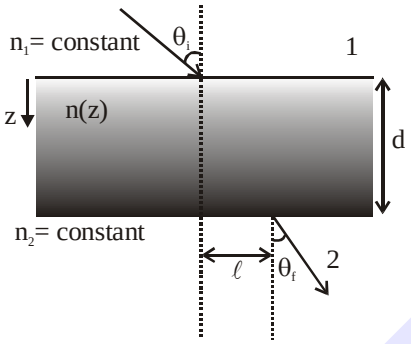
(A) 15° (B) 22.5°

(C) 30° (D) 45°

GO0216

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13. A transparent slab of thickness d has a refractive index n(z) that increases with z. Here z is the vertical distance inside the slab, measured from the top. The slab is placed between two media with uniform refractive indices n_1 and n_2 ($> n_1$), as shown in the figure. A ray of light is incident with angle θ_i from medium 1 and emerges in medium 2 with refraction angle θ_f with a lateral displacement ℓ. Which of the following statement(s) is(are) true ? **[JEE-Advance-2016]**



(A) ℓ is independent of n_2
(B) ℓ is dependent on n(z)
(C) $n_1\sin\theta_i = (n_2 - n_1)\sin\theta_f$
(D) $n_1\sin\theta_i = n_2\sin\theta_f$

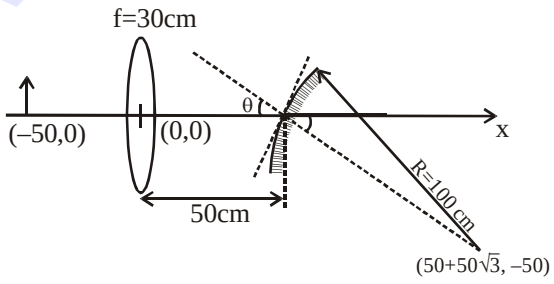
GO0217

14. A plano-convex lens is made of a material of refractive index n. When a small object is placed 30 cm away in front of the curved surface of the lens, an image of double the size of the object is produced. Due to reflection from the convex surface of the lens, another faint image is observed at a distance of 10 cm away from the lens. Which of the following statement(s) is(are) true? **[JEE-Advance-2016]**

(A) The refractive index of the lens is 2.5
(B) The radius of curvature of the convex surface is 45 cm
(C) The faint image is erect and real
(D) The focal length of the lens is 20 cm.

GO0218

15. A small object is placed 50 cm to the left of thin convex lens of focal length 30 cm. A convex spherical mirror of radius of curvature 100 cm is placed to the right of the lens at a distance of 50 cm. The mirror is tilted such that the axis of the mirror is at an angle $\theta = 30°$ to the axis of the lens, as shown in the figure. If the origin of the coordinate system is taken to be at the centre of the lens, the coordinates (in cm) of the point (x, y) at which the image is formed are : **[JEE-Advance-2016]**



(A) $(25, 25\sqrt{3})$ (B) $\left(\frac{125}{3}, \frac{25}{\sqrt{3}}\right)$ (C) $(50 - 25\sqrt{3}, 25)$ (D) (0, 0)

GO0219

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16. For an isosceles prism of angle A and refractive index μ, it is found that the angle of minimum deviation $\delta_m = A$. Which of the following options is/are correct ? **[JEE-Advance-2017]**

(A) At minimum deviation, the incident angle i_1 and the refracting angle r_1 at the first refracting surface are related by $r_1 = (i_1/2)$

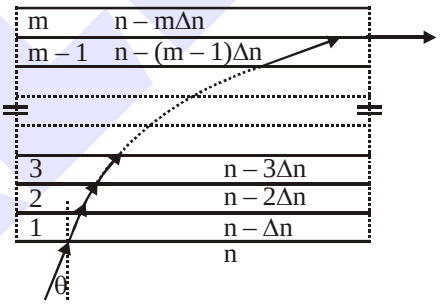
(B) For this prism, the refractive index μ and the angle of prism A are related as $A = \frac{1}{2}\cos^{-1}\left(\frac{\mu}{2}\right)$

(C) For this prism, the emergent ray at the second surface will be tangential to the surface when the angle of incidence at the first surface is $i_1 = \sin^{-1}\left[\sin A\sqrt{4\cos^2\frac{A}{2}-1}-\cos A\right]$

(D) For the angle of incidence $i_1 = A$, the ray inside the prism is parallel to the base of the prism.

GO0220

17. A monochromatic light is travelling in a medium of refractive index n = 1.6. It enters a stack of glass layers from the bottom side at an angle θ = 30°. The interfaces of the glass layers are parallel to each other. The refractive indices of different glass layers are monotonically decreasing as $n_m = n - m\Delta n$, where n_m is the refractive index of the mth slab and Δn = 0.1 (see the figure). The ray is refracted out parallel to the interface between the $(m-1)^{th}$ and m^{th} slabs from the right side of the stack. What is the value of m ? **[JEE-Advance-2017]**



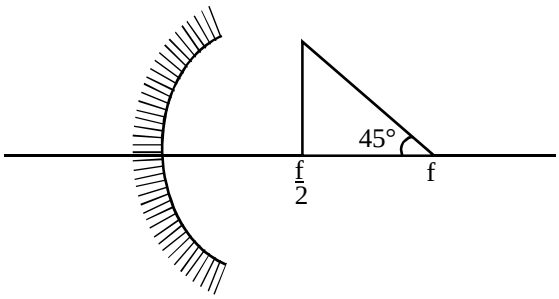
GO0221

18. Sunlight of intensity 1.3 kW m^{-2} is incident normally on a thin convex lens of focal length 20 cm. Ignore the energy loss of light due to the lens and assume that the lens aperture size is much smaller than its focal length. The average intensity of light, in kW m^{-2}, at a distance 22 cm from the lens on the other side is _____. **[JEE-Advance-2018]**

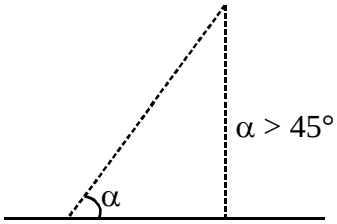
GO0222

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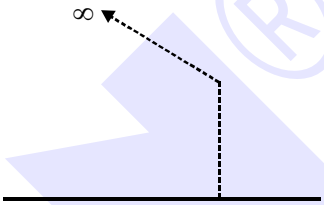
19. A wire is bent in the shape of a right angled triangle and is placed in front of a concave mirror of focal length f, as shown in the figure. Which of the figures shown in the four options qualitatively represent(s) the shape of the image of the bent wire ? (These figures are not to scale.) ? **[JEE-Advance-2018]**



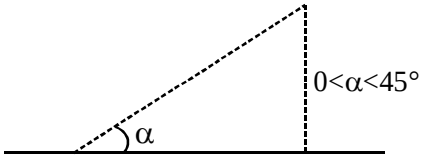
(A)



(B)



(C)

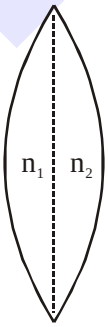


(D)



GO0223

20. A thin convex lens is made of two materials with refractive indices n_1 and n_2, as shown in figure. The radius of curvature of the left and right spherical surfaces are equal. f is the focal length of the lens when $n_1 = n_2 = n$. The focal length is $f + \Delta f$ when $n_1 = n$ and $n_2 = n + \Delta n$. Assuming $\Delta n << (n-1)$ and $1 < n < 2$, the correct statement(s) is/are : **[JEE-Advance-2019]**



(1) The relation between $\frac{\Delta f}{f}$ and $\frac{\Delta n}{n}$ remains unchanged if both the convex surfaces are replaced by concave surfaces of the same radius of curvature.

(2) $\left|\frac{\Delta f}{f}\right| < \left|\frac{\Delta n}{n}\right|$

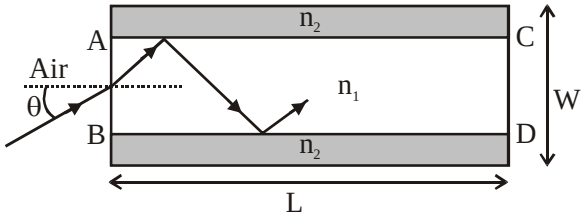
(3) For $n = 1.5$, $\Delta n = 10^{-3}$ and $f = 20$ cm, the value of $|\Delta f|$ will be 0.02 cm (round off to 2nd decimal place)

(4) If $\frac{\Delta n}{n} < 0$ then $\frac{\Delta f}{f} > 0$

GO0224

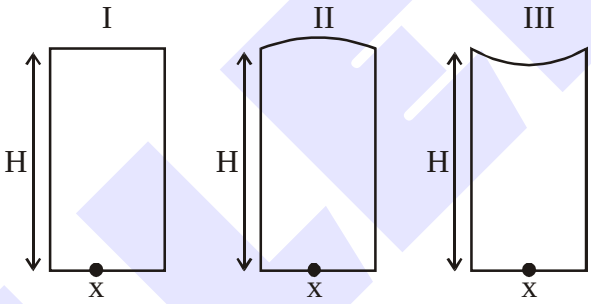
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21. A planar structure of length L and width W is made of two different optical media of refractive indices $n_1 = 1.5$ and $n_2 = 1.44$ as shown in figure. If L >> W, a ray entering from end AB will emerge from end CD only if the total internal reflection condition is met inside the structure. For L = 9.6 m, if the incident angle θ is varied, the maximum time taken by a ray to exit the plane CD is $t \times 10^{-9}$ s, where t is ____. [Speed of light $c = 3 \times 10^8$ m/s] **[JEE-Advance-2019]**



GO0225

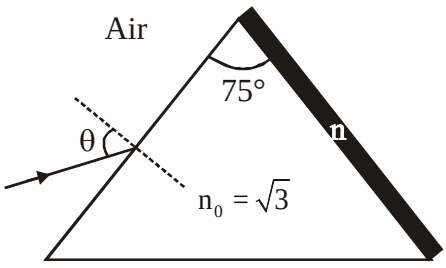
22. Three glass cylinders of equal height H = 30 cm and same refractive index n = 1.5 are placed on a horizontal surfaces shown in figure. Cylinder I has a flat top, cylinder II has a convex top and cylinder III has a concave top. The radii of curvature of the two curved tops are same (R = 3m). If H_1, H_2 and H_3 are the apparent depths of a point X on the bottom of the three cylinders, respectively, the correct statement(s) is/are **[JEE-Advance-2019]**



(1) $H_3 > H_1$
(2) $0.8 \text{ cm} < (H_2 - H_1) < 0.9 \text{ cm}$
(3) $H_2 > H_3$
(4) $H_2 > H_1$

GO0226

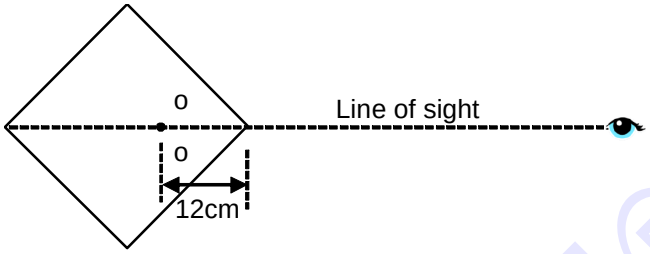
23. A monochromatic light is incident from air on a refracting surface of a prism of angle 75° and refractive index $n_0 = \sqrt{3}$. The other refracting surface of a prism is coated by a thin film of material of refractive index n as shown in figure. The light suffers total internal reflection at the coated prism surface for an incidence angle of $\theta \le 60°$. The value of n^2 is_____. **[JEE-Advance-2019]**



GO0227

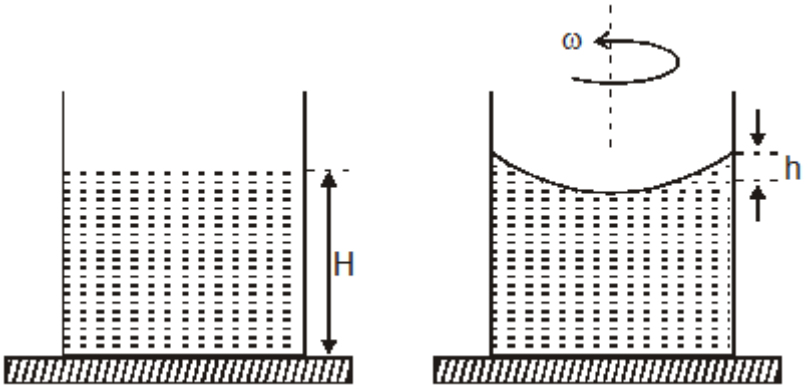
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24. A large square container with thin transparent vertical walls and filled with water (refractive index $\frac{4}{3}$) is kept on a horizontal table. A student holds a thin straight wire vertically inside the water 12 cm from one of its corners, as shown schematically in the figure. Looking at the wire from this corner, another student sees two images of the wire, located symmetrically on each side of the line of sight as shown. The separation (in cm) between these images is _____. **[JEE-Advance-2020]**



GO0266

25. A beaker of radius r is filled with water (refractive index $\frac{4}{3}$) up to a height H as shown in the figure on the left. The beaker is kept on a horizontal table rotating with angular speed ω. This makes the water surface curved so that the difference in the height of water level at the center and at the circumference of the beaker is h ($h << H$, $h << r$), as shown in the figure on the right. Take this surface to be approximately spherical with a radius of curvature R. Which of the following is/are correct? (g is the acceleration due to gravity) **[JEE-Advance-2020]**



(A) $R = \frac{h^2 + r^2}{2h}$

(B) $R = \frac{3r^2}{2h}$

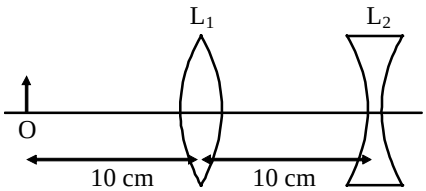
(C) Apparent depth of the bottom of the beaker is close to $\frac{3H}{2}\left(1+\frac{\omega^2 H}{2g}\right)^{-1}$

(D) Apparent depth of the bottom of the beaker is close to $\frac{3H}{4}\left(1+\frac{\omega^2 H}{4g}\right)^{-1}$

GO0267

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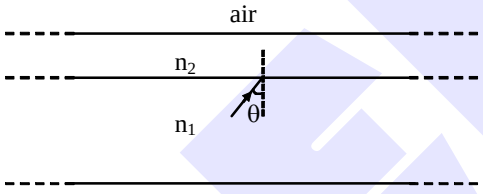
26. An extended object is placed at point O, 10 cm in front of a convex lens L_1 and a concave lens L_2 is placed 10 cm behind it, as shown in the figure. The radii of curvature of all the curved surfaces in both the lenses are 20 cm. The refractive index of both the lenses is 1.5. The total magnification of this lens system is: **[JEE-Advance-2021]**



(A) 0.4 (B) 0.8 (C) 1.3 (D) 1.6

GO0268

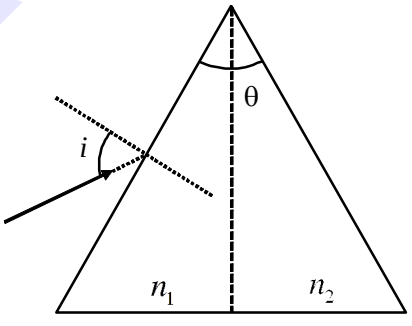
27. A wide slab consisting of two media of refractive indices n_1 and n_2 is placed in air as shown in the figure. A ray of light is incident from medium n_1 to n_2 at an angle θ, where $\sin\theta$ is slightly larger than $1/n_1$. Take refractive index of air as 1. Which of the following statement(s) is(are) correct? **[JEE-Advance-2021]**



(A) The light ray enters air if $n_2 = n_1$.
(B) The light ray is finally reflected back into the medium of refractive index n_1 if $n_2 < n_1$.
(C) The light ray is finally reflected back into the medium of refractive index n_1 if $n_2 > n_1$.
(D) The light ray is reflected back into the medium of refractive index n_1 if $n_2 = 1$.

GO0269

28. For a prism of prism angle $\theta = 60^o$, the refractive indices of the left half and the right half are, respectively, n_1 and n_2 ($n_2 \geq n_1$) as shown in the figure. The angle of incidence i is chosen such that the incident light rays will have minimum deviation if $n_1 = n_2 = n = 1.5$. For the case of unequal refractive indices, $n_1 = n$ and $n_2 = n + \Delta n$ (where $\Delta n << n$), the angle of emergence $e = i + \Delta e$. Which of the following statement(s) is (are) correct ? **[JEE-Advance-2021]**



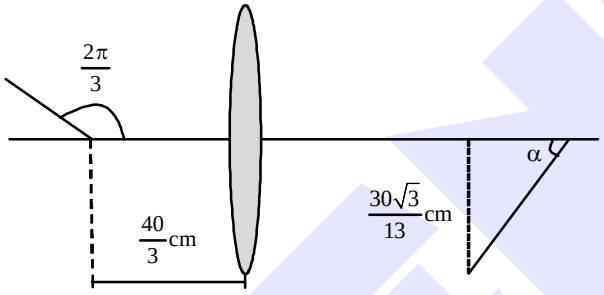
(A) The value of Δe (in radians) is greater than that of Δn.
(B) Δe is proportional to Δn.
(C) Δe lies between 2.0 and 3.0 milliradians, if $\Delta n = 2.8 \times 10^{-3}$.
(D) Δe lies between 1.0 and 1.6 milliradians, if $\Delta n = 2.8 \times 10^{-3}$.

GO0270

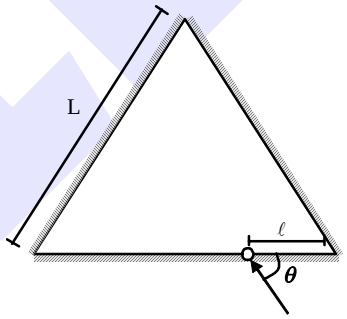
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29. A rod of length 2 cm makes an angle $\frac{2\pi}{3}$ rad with the principal axis of a thin convex lens. The lens has a focal length of 10 cm and is placed at a distance of $\frac{40}{3}$ cm from the object as shown in the figure. The height of the image is $\frac{30\sqrt{3}}{13}$ cm and the angle made by it with respect to the principal axis is α rad. The value of α is $\frac{\pi}{n}$ rad, where n is _____. **[JEE-Advance-2022]**

GO0416



30. Three plane mirrors form an equilateral triangle with each side of length L. There is a small hole at a distance $\ell > 0$ from one of the corners as shown in the figure. A ray of light is passed through the hole at an angle θ and can only come out through the same hole. The cross section of the mirror configuration and the ray of light lie on the same plane. **[JEE-Advance-2022]**



Which of the following statement(s) is(are) correct? **GO0417**

(A) The ray of light will come out for $\theta = 30°$, for $0 < \ell < L$.

(B) There is an angle for $l = \frac{L}{2}$ at which the ray of light will come out after two reflections.

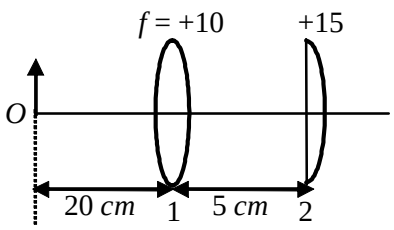
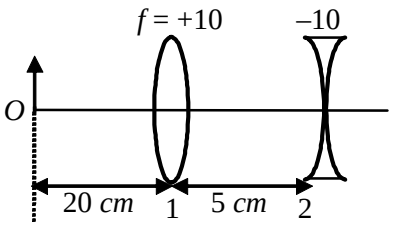
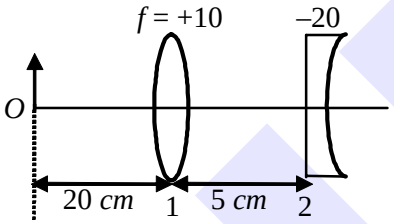
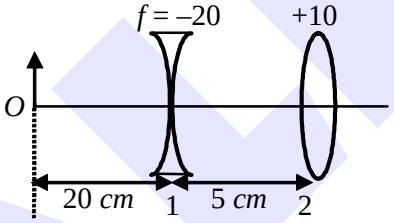
(C) The ray of light will NEVER come out for $\theta = 60°$, and $l = \frac{L}{3}$.

(D) The ray of light will come out for $\theta = 60°$, and $0 < l < \frac{L}{2}$ after six reflections.

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31. List I contains four combinations of two lenses (1 and 2) whose focal lengths (in cm) are indicated in the figures. In all cases, the object is placed 20 cm from the first lens on the left, and the distance between the two lenses is 5 cm. List II contains the positions of the final images. **GO0418**

[JEE-Advance-2022]

List-I		**List-II**	
(I)		(P)	Final image is farmed at 7.5 *cm* on the right side of lens 2.
(II)		(Q)	Final image is formed at 60.0 *cm* on the right side of lens 2.
(III)		(R)	Final image is formed at 30.0 *cm* on the left side of lens 2.
(IV)		(S)	Final image is formed at 6.0 *cm* on the right side of lens 2.
		(T)	Final image is formed at 30.0 *cm* on the right side of lens 2.

Which one of the following options is correct ?

(A) I → P, II → R, III → Q, IV → T

(B) I → Q, II → P, III → T, IV → S

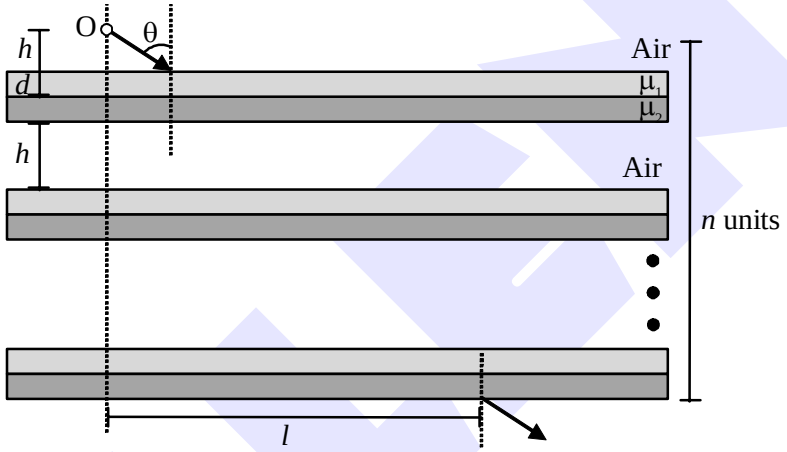
(C) I → P, II → T, III → R, IV → Q

(D) I → T, II → S, III → Q, IV → R

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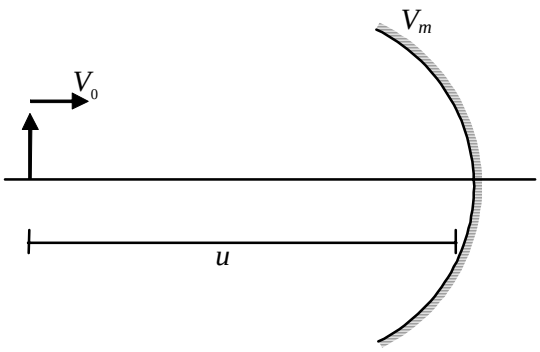
32. Consider a configuration of n identical units, each consisting of three layers. The first layer is a column of air of height h = $\frac{1}{3}$ cm, and the second and third layers are of equal thickness $d = \frac{\sqrt{3}-1}{2} cm$, and refractive indices $\mu_1 = \sqrt{\frac{3}{2}}$ and $\mu_2 = \sqrt{3}$, respectively. A light source O is placed on the top of the first unit, as shown in the figure. A ray of light from O is incident on the second layer of the first unit at an angle of θ = 60° to the normal. For a specific value of n, the ray of light emerges from the bottom of the configuration at a distance $l = \frac{8}{\sqrt{3}}$ cm, as shown in the figure. The value of n is _____.

[JEE-Advance-2022]
GO0419



33. An object and a concave mirror of focal length f = 10 cm both move along the principal axis of the mirror with constant speeds. The object moves with speed V_0 = 15 cm s^{-1} towards the mirror with respect to a laboratory frame. The distance between the object and the mirror at a given moment is denoted by u. When u = 30 cm, the speed of the mirror V_m is such that the image is instantaneously at rest with respect to the laboratory frame, and the object forms a real image. The magnitude of V_m is _____ cm s^{-1}.

[JEE-Advance-2022]
GO0420



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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	C	D	A	D	B	C	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	D	C	B	B	C	B	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	B	A	B	B	C	B	A	C	C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	C	C	A	D	C	C	B	D	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	A	C	A	C	B	C	B	C	C	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	A	C	C	C	B	B	B	C	A	A
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	B	D	A	D	B	A	B	B	A	C
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	D	B	D	B	C	B	A	C	A	C
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	A	D	D	C	B	B	B	B	B	B
Que.	91	92	93	94	95	96	97	98	99	100
Ans.	A	D	D	B	D	A	D	C	A	C
Que.	101	102	103							
Ans.	C	B	B							

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EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	C	D	D	B	B	D	C	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	D	A	B	D	D	B	C	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	C	D	C	A	C	B	A	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	B	A	C	B	B	B	B	C	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	B	A	A	B,C	A,D	B,C	A,B,C	B,D	A,C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	A,C	A,D	A,C,D	B,C	A,B	A,B,C,D	A,C	A,D	B,C,D	A,B,C
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	B	B	C	C	B	D	C	B	D	D
Que.	71	A	B	C	D	72	A	B	C	D
Ans.		Q	P	S	R		Q	P	S	R
Que.	73	74	A	B	C	D				
Ans.	B		P,R	Q,S,T	P,R,T	Q,S				
Que.	75	A	B	C	D					
Ans.		P	Q	R,T	P,T					

EXERCISE # S

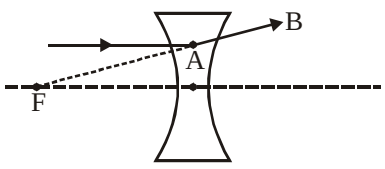
1. 3m **2.** 2 **3.** 4 **4.** 2 **5.** 25 **6.** 4 **7.** 1 **8.** 5
9. d/2 **10.** 16 mm^3 **11.** 4 **12.** 5 **13.** 5 **14.** 4 **15.** 8

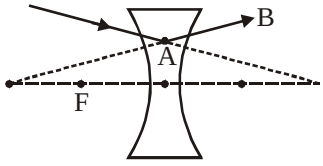
16. 42 cm **17.** (a) $\sin^{-1}\left(\frac{1}{5}\right)$ (b) air **18.** $\theta < \sin^{-1}(2\sin 15°)$

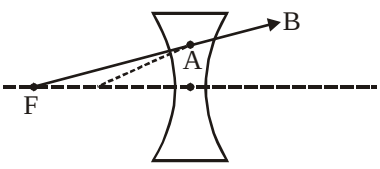
19. $\left(\frac{8}{3}mm\right)$ **20.** 2 **21.** 9 **22.** 8 **23.** 5cm **24.** $f = v = \frac{\mu_3 R}{\mu_3 - \mu_1}$

25. (i) 0.2 m, (ii) 0.4 m **26.** 3 **27.** 5 **28.** $y = -\frac{x}{1200} + 0.1$ **29.** 30

30. Real, below principal axis, anywhere b/w P & ∞

31. (1) 

(2) 

(3) 

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32. $(\pi/4)$ cm^2 **33.** 17.5 cm **34.** 2 **35.** (–20 cm) **36.** 3 **37.** 1 **38.** 4

39. (i) $\lambda_0 = 600$ nm, n = 1.5 (ii) $i = \sin^{-1}(0.75) = 48.59^o$

40. (a) f = –20 cm, (b) + 80 cm, convergent achromatic lens] **41.** 8 X **42.** 90 cm, 9

43. 3 cm **44.** + 60 D to + 51 D

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	4	4	3	1	2	3	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	1	2	2	2	4	1	1	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	158	By NTA (5) By Allen (476)	By NTA (5) By Allen (4.48)	4	By NTA (3) By Allen (2)	3	12	2	25
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	3	50	1	2	4	12	210	A	A	B
Que.	41	42	43							
Ans.	10	27	15							

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	D	A,C	C	B	7	B	2	A,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	A	A,B,D	A,D	A	A,C,D	8	130 [129.95, 130.05]	D	1,3,4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	50.00	3,4	1.50	Bonus	A,D	B	B,C,D	B,C	6	A,B
Que.	31	32	33							
Ans.	A	4	3							

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